MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

n waves are produced on a string in 1 s. When the radius of the string is doubled and the tension is maintained the same, the number of waves produced in 1 s for the same harmonic will be

(A) $2n$

(B) $\frac{n}{3}$

(C) $\frac{n}{2}$

(D) $\frac{n}{\sqrt{2}}$

Q.2

Match the following

	Column I		Column II
(a)	Enthalpy	(p)	Intensive property
(b)	Temperature	(q)	Path function
(c)	Free energy	(r)	Function of U, P and V
(d)	Work	(s)	State function

(A) $a \rightarrow r, s\ , b \rightarrow p, s\ , c \rightarrow r, s\ , d \rightarrow q$

(B) $a \rightarrow p\ , b \rightarrow r\ , c \rightarrow q\ , d \rightarrow r$

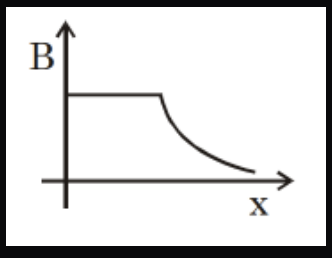
(C) $a \rightarrow r, b \rightarrow q, c \rightarrow r, d \rightarrow p$

(D) $a \rightarrow q, b \rightarrow r, c \rightarrow p\ , d \rightarrow r$

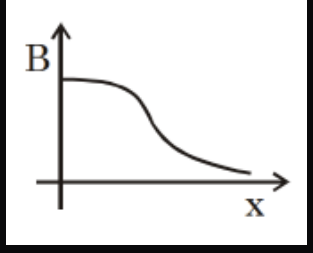
Q.3

An infinite cylindrical current carrying wire carries a current that is uniformly distributed over its cross section. The magnetic field as a function of distance from it's axis is :-

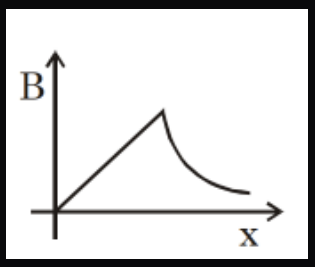
(A)



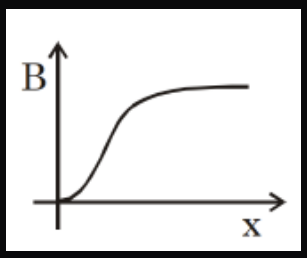
(B)



(C)



(D)



Q.4

A car is moving on a circular road of radius 150 m with a coefficient of friction 0.6. With what optimum velocity should the car be moved so that it crosses the turn safely.:

(A) $60\ \mathrm{ms}^{-1}$

(B) $30\ ms^{-1}$

(C) $15\ ms^{-1}$

(D) $45\ \mathrm{ms}^{-1}$

Q.5

A satellite moves round the earth in a circular orbit of radius R making one revolution per day. A second satellite moving in a circular orbit, moves round the earth once in 8 days. The radius of the orbit of the second satellite is

(A) $8\mathrm{R}$

(B) $4\mathrm{R}$

(C) $2\mathrm{R}$

(D) R

Q.6

Six wires each of resistance r, are connected so as to form a tetrahedron. The equivalent resistance of the combination when current enters through one corner and leave through some other corner is

(A) r

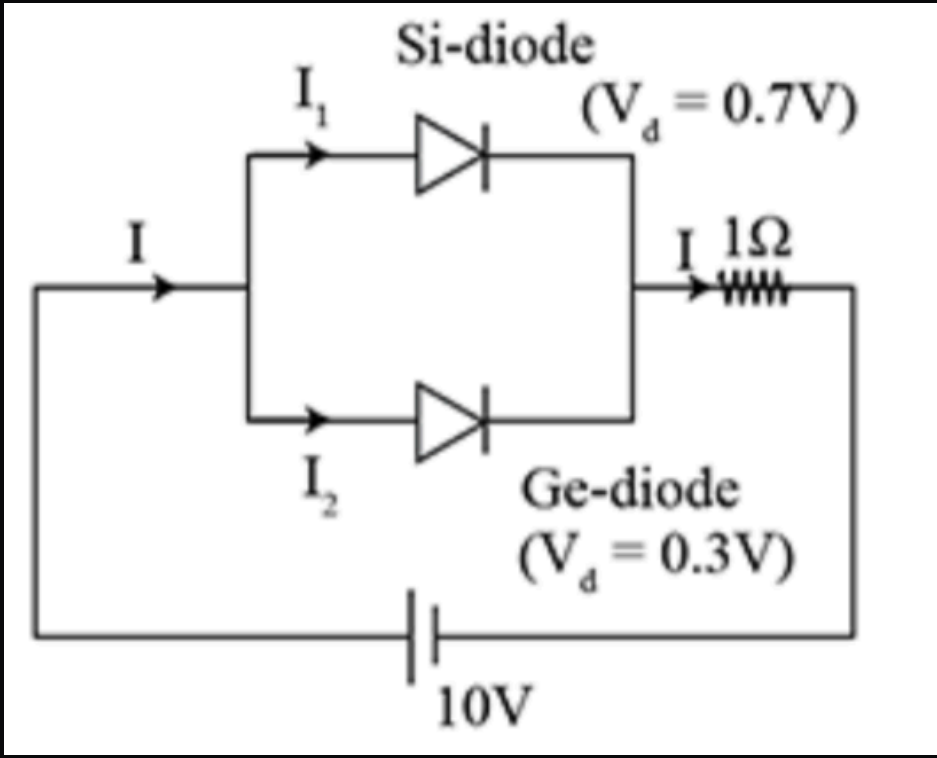
(B) $2r$

(C) $\frac{r}{3}$

(D) $\frac{r}{2}$

Q.7

In the given circuit, which of following option is correct?



(A) $I_1 = 10A$

(B) $I_2 = 9.7\ \text{A}$

(C) $I = 10\ \text{A}$

(D) $I_1 = 9.3\ \text{A}$

Q.8

Two seconds after projection, a projectile is traveling in a direction inclined at 30° to the horizontal. After one more sec, it is traveling horizontally, the magnitude and direction of its velocity are

(A) $2\sqrt{2}\ \text{m s}^{-1},\ 60^\circ$

(B) $20\sqrt{3}\ \text{m s}^{-1},\ 60^\circ$

(C) $6\sqrt{40}\ \text{m s}^{-1},\ 30^\circ$

(D) $40\sqrt{6}\ \text{m s}^{-1},\ 30^\circ$

Q.9

Assertion: When two soap bubbles having different radii are kept in contact, the common surface at their interface will bulge into a large bubble.
Reason: Pressure inside the smaller bubble is larger.

(A) Assertion is true, reason is true, reason is correct explanation for assertion.

(B) Assertion is true, reason is true, reason is not a correct explanation for assertion.

(C) Assertion is true, reason is false.

(D) Assertion is false, reason is true.

Q.10

The amplitude of an electromagnetic wave in vacuum is doubled with no other changes made to the wave. As a result of this doubling of the amplitude, which of the following statement is correct?

(A) Only the speed of wave propagation changes.

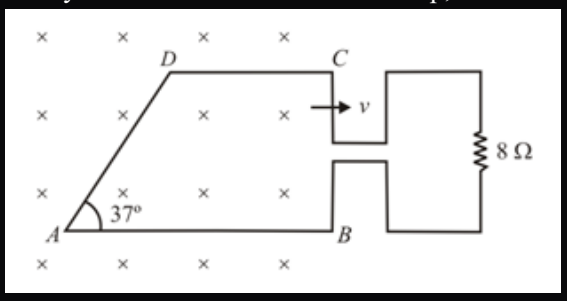
(B) Only the frequency of the wave changes.

(C) Only the wavelength of the wave changes.

(D) None of these.

Q.11

The loop $ABCD$ is moving with velocity v towards right. The magnetic field is 4 T. The loop is connected to a resistance of 8 Ω. If steady current of 2 A flows in the loop, then value of v, if loop has resistance of 4 Ω, is (given, $AB = 30\ \text{cm}$, $AD = 30\ \text{cm}$)



(A) $20\ \text{m s}^{-1}$

(B) $10\ \text{m s}^{-1}$

(C) $\frac{100}{3}\ \text{m s}^{-1}$

(D) $\frac{10}{3}\ \text{m s}^{-1}$

Q.12

Assertion: For a point light source, frequency of light emitted is numerically equal to number of photons emitted per second.

Reason: According to wave theory frequency of light wave is equivalent to number of cycles completed per second and wave-particle duality explains the dual nature of electromagnetic waves as a wave or as a corpuscle.

(A) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

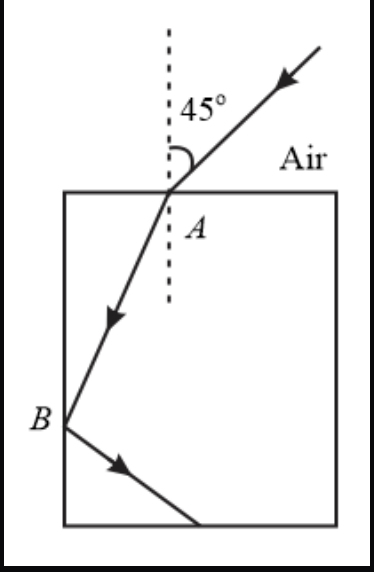
(B) If both Assertion and Reason are true but Reason is not explanation of the Assertion.

(C) If Assertion is true but the Reason is false.

(D) If Assertion is false but Reason is true.

Q.13

A ray of light is incident at an angle of 45° from air to a glass slab at point A, the angle of incidence being 45°.The minimum value of the refractive index of the material of the slab so that Total Internal Reflection takes place at B is



(A) $\sqrt{\frac{3}{2}}$

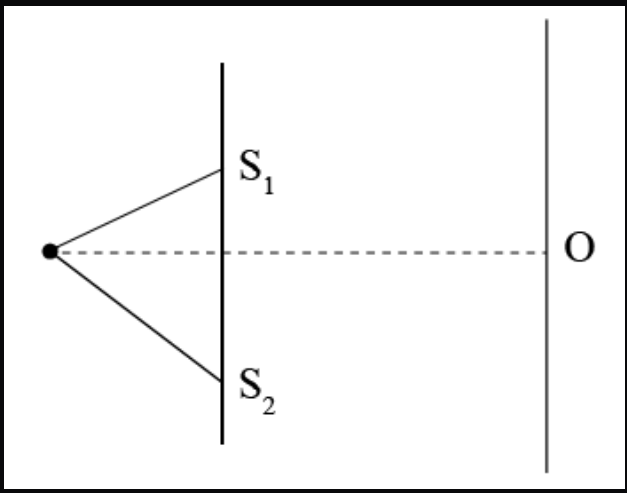
(B) $\sqrt{3}$

(C) $\sqrt{\frac{4}{3}}$

(D) None of these

Q.14

In the setup shown in figure, the two slits, S_1, and S_2 are not equidistant from the slit S. The central fringe at O is, then



(A) always bright

(B) always dark

(C) Either dark or bright depending on the position of S

(D) Neither dark nor bright

Q.15

If $E =$ energy, $G =$ gravitational constant, $I =$ impulse and $M =$ mass, then dimensions of $\frac{GIM^2}{E^2}$ are same as that of

(A) Time

(B) Mass

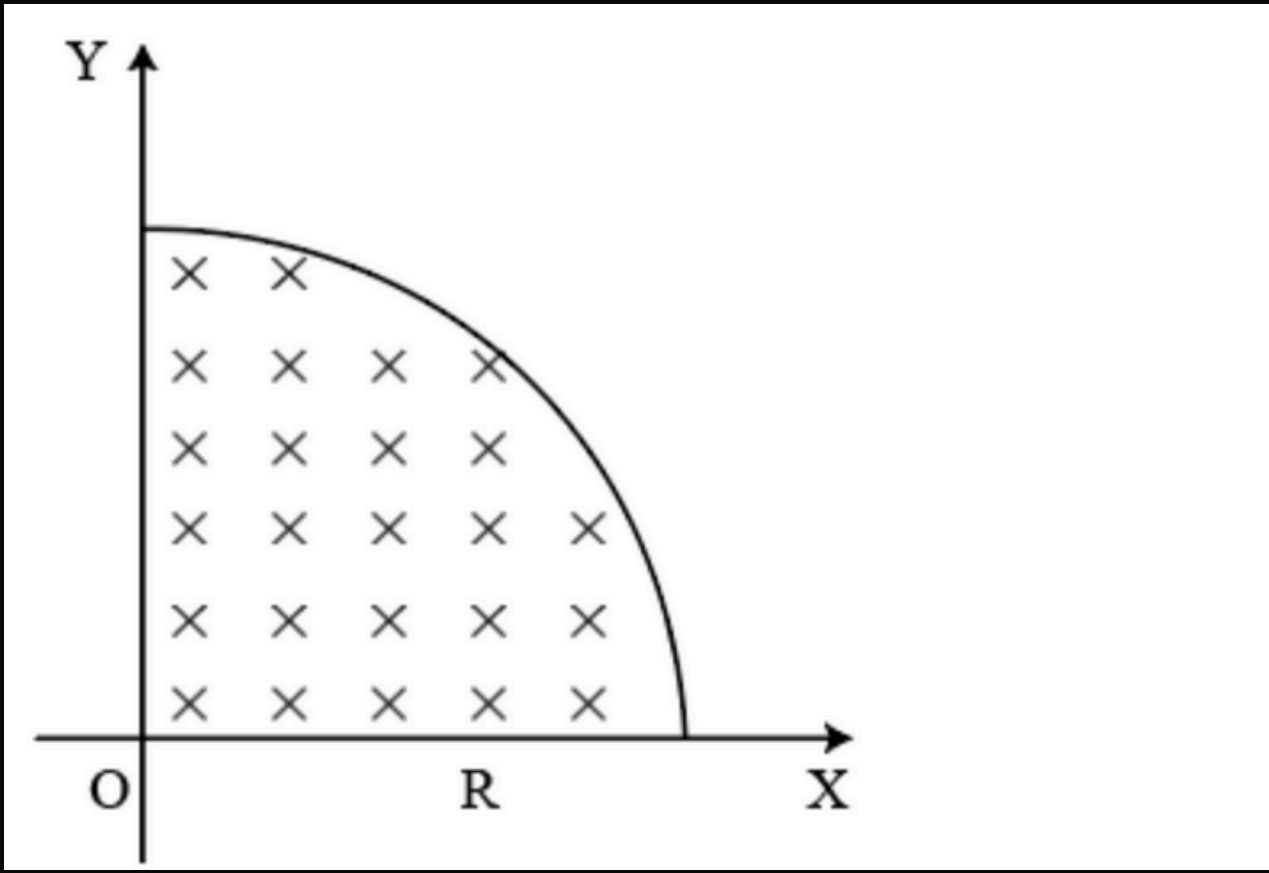
(C) Length

(D) Force

Q.16

The quarter disc of radius R (see figure) has a uniform surface charge density σ.

Find electric potential at a point (O, O, Z).



(A) $\frac{\sigma}{8\epsilon_0}\left[1-\frac{Z}{\sqrt{R^2+Z^2}}\right]$

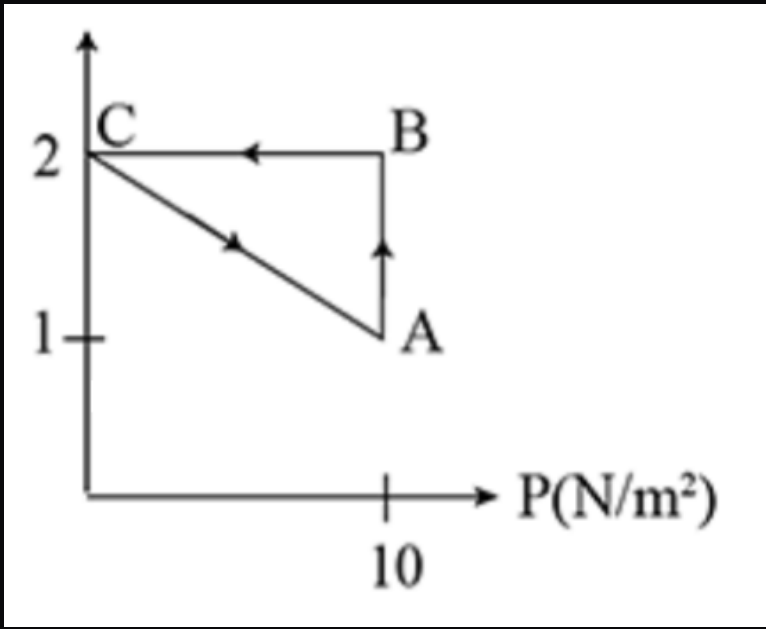
(B) $\frac{\sigma}{4\epsilon_0}\left[\sqrt{R^2+Z^2}+Z\right]$

(C) $\frac{\sigma}{8\epsilon_0}\left[\sqrt{R^2+Z^2}-Z\right]$

(D) $\frac{\sigma}{8\epsilon_0}\left[1+\frac{Z}{\sqrt{R^2+Z^2}}\right]$

Q.17

An ideal gas is taken through a cycle $A \rightarrow B \rightarrow C \rightarrow A$, as $V\left(m^3\right)$ shown in the figure. If net heat supplied to the gas in the cycle is $5J$, the work done by gas in the process $C-A$ is:



(A) $-5J$

(B) $-10\ \mathrm{J}$

(C) $-15J$

(D) $-20\ \mathrm{J}$

Q.18

Using the following data
Mass of hydrogen atom $= 1.00783u$
Mass of neutron $= 1.00867\mathrm{u}$
Mass of nitrogen atom $\left({}_7\mathrm{N}^{14}\right) = 14.00307\mathrm{u}$
The calculated value of the binding energy of the nucleus of the nitrogen atom $\left({}_7\mathrm{N}^{14}\right)$ is close to

(A) $56\mathrm{MeV}$

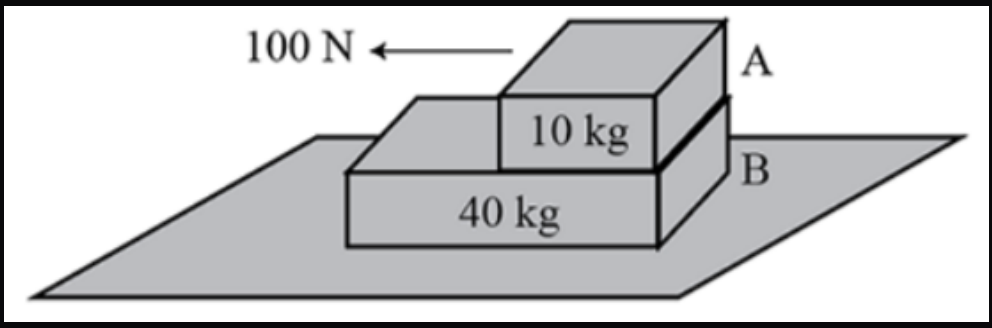
(B) $98\mathrm{MeV}$

(C) $104\mathrm{MeV}$

(D) $112\mathrm{MeV}$

Q.19

A $40\ \mathrm{kg}$ slab rests on a frictionless floor as shown in the figure. A $10\ \mathrm{kg}$ block rests on the top of the slab. The static coefficient of friction between the block and slab is 0.60 while the coefficient of kinetic friction is 0.40. The $10\ \mathrm{kg}$ block is acted upon by a horizontal force $100\ \mathrm{N}$. If $\mathrm{g} = 9.8\ \mathrm{m/s^2}$, the resultaing acceleration of the slab will be



(A) $0.98\ \mathrm{m/s^2}$

(B) $1.47\ \mathrm{m/s^2}$

(C) $1.52\ \mathrm{m/s^2}$

(D) $6.1\ \mathrm{m/s^2}$

Q.20

A proton and an electron enter into a magnetic field $\mathbf{B}$ perpendicular to it with speed v and $10v$. Then, the ratio of the radii of their circular paths is (given that, proton is 1820 times heavier than electron)

(A) $1820 : 1$

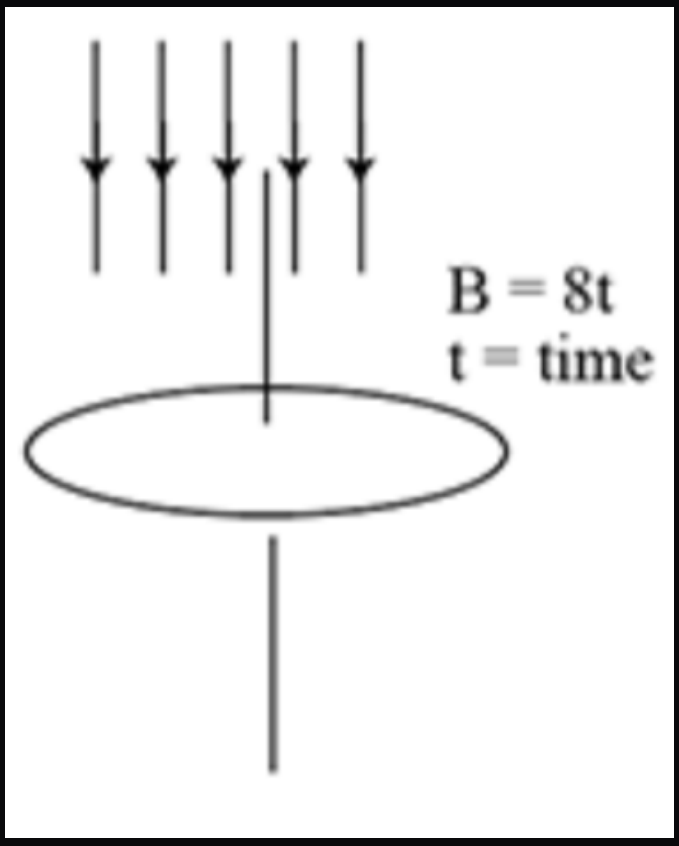
(B) 18200: 1

(C) 1: 182

(D) $182 : 1$

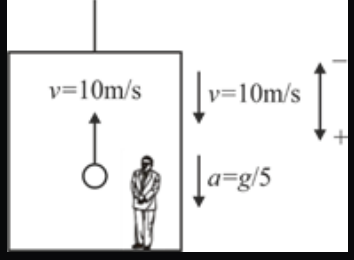
Q.21

A thin ring of mass $0.1\ \mathrm{kg}$, and radius $0.5\ \mathrm{m}$ carrying charge $8C$ is free to rotate about an axis on horizontal frictionless surface as shown in the figure. A variable magnetic field of $8t$ tesla is applied vertically downward on the ring. If ring is at rest before applying magnetic field, the kinetic energy acquired by ring after $10\ \mathrm{s}$ is: (answer in KJ)



Q.22

A boy is in a lift which is moving downwards with an acceleration $g/5$. At an instant when the velocity of the lift is $10\ \mathrm{m/s}$, then boy throws a ball upwards with a velocity $10\ \mathrm{m/s}$. The time after which ball will return into the hand of boy is (take $g = 10\ m/s^2$)



Q.23

A ball of mass $0.2\ \text{kg}$ is thrown vertically upwards by applying a force by hand. If the hand moves $0.2\ \text{m}$. while applying the force and the ball goes upto $2\ \text{m}$ height further. Find the magnitude of force (Consider $g = 10\ \text{m/s}^2$)

Q.24

A large horizontal surface moves up and down in SHM with an amplitude of $1\ \text{cm}$. If a mass of $10\ \text{kg}$ (which is placed on the surface) is to remain continuously. in contact with it, the maximum frequency of SHM will be ...Hz. (Mark answer to nearest whole number) (given: $g = 9.8\ \text{m/s}^2$ and $\pi = 22/7$)

Q.25

How much heat energy (in kcal) is necessary to raise the temperature of 5 kg of water from $20°C$ to $100°C$?

Q.26

$[\text{PdCl}_2(\text{PMe}_3)_2]$ is a diamagnetic complex of $\text{Pd}(II)$. How many total isomers are possible for $[\text{NiCl}_2(\text{PMe}_3)_2]$ having oxidation state of Ni is (II) ? (do not count given compound in isomers)

(A) 3

(B) 2

(C) 1

(D) Zero

Q.27

Place the following alcohols in decreasing order of rate of dehydration with concentration H_2SO_4 .

1. $CH_3CH_2CH(OH)CH_2CH_2CH_3$
2. $(CH_3)_2C(OH)CH_2CH_2CH_3$
3. $(CH_3)_2C(OH)CH(CH_3)_2$
4. $CH_3CH_2CH(OH)CH(CH_3)_2$
5. $CH_3CH_2CH_2CH_2CH_2CH_2OH$

(A) 3 > 2 > 4 > 5 > 1

(B) 3 > 2 > 4 > 1 > 5

(C) 3 > 2 > 1 > 4 > 5

(D) 3 > 2 > 1 > 5 > 4

Q.28

When dilute H_2SO_4 and H_2O_2 are added to a solution of chromate ions, an intense blue colour is produced, which is stable in ether. This is due to the formation of

(A) $Cr_2O_7^{2-}$

(B) Cr_2O_3

(C) CrO_5

(D) CrO_3

Q.29

Consider two reactions having same Arrhenius factor A, but different energy of activation.

(i) $A \rightarrow B; Ea_1 = 20\ kJ$

(ii) $C \rightarrow D; Ea_2 = 30\ kJ$

Both are at temperature $25^\circ C$ If temperature in both reaction is increased slightly in such a way that change in temperature in both case is same than choose the correct options.

(A) The second reaction is faster

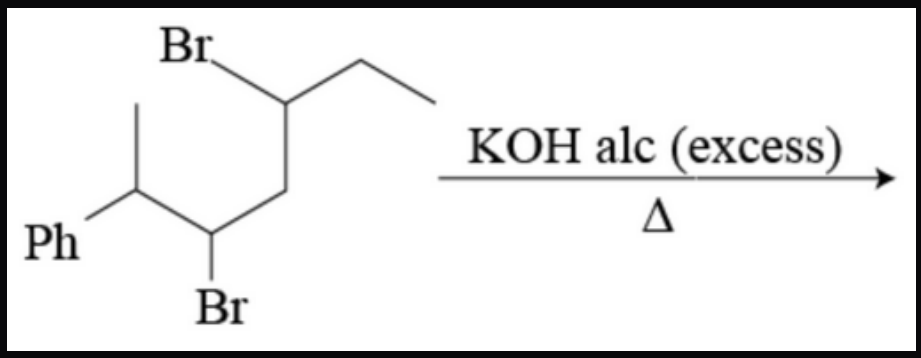
(B) The second reaction is more sensitive towards temperature variation

(C) If temperature increases, rate of first reaction increase more sharply

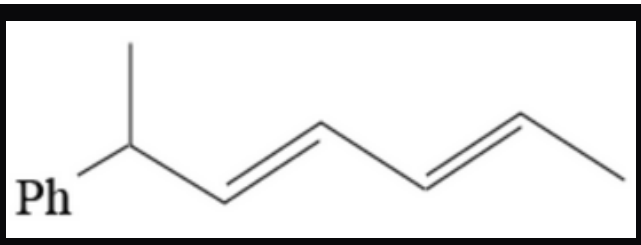
(D) All the above are correct

Q.30

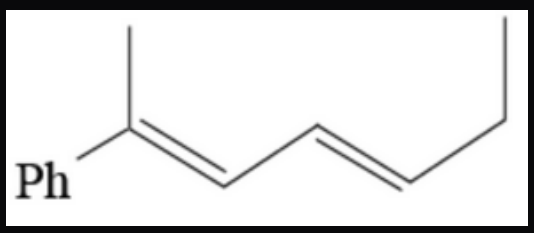
The major product of the following reaction is



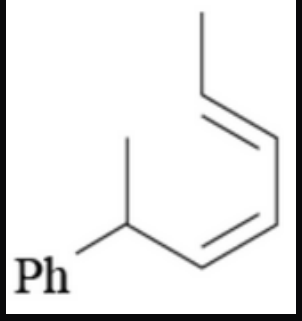
(A)



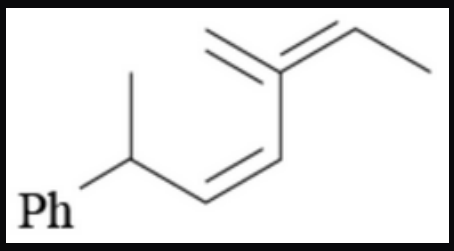
(B)



(C)



(D)



Q.31

Identify correct statements about NO molecule:

(i) When NO is ionized to NO^+, the electron is removed from the π^*2p orbital.

(ii) Bond order of NO is 2.5 and bond order of NO^+ is 3.

(iii) Bond length of NO^+ is greater than that of NO.

(iv) It is similar to N_2^- in all respect.

(A) (i) and (ii)

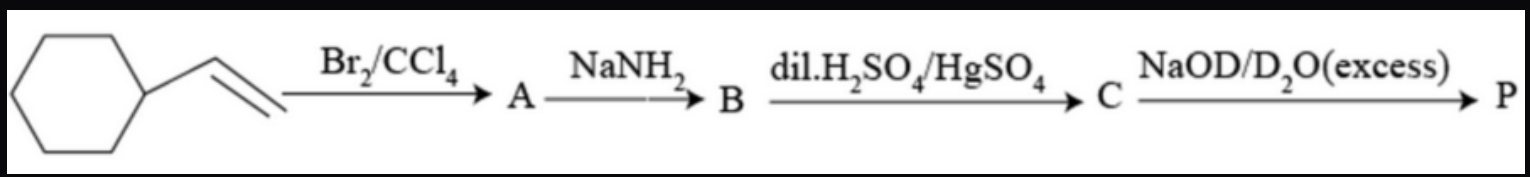
(B) (i) and (iii)

(C) (i) and (iv)

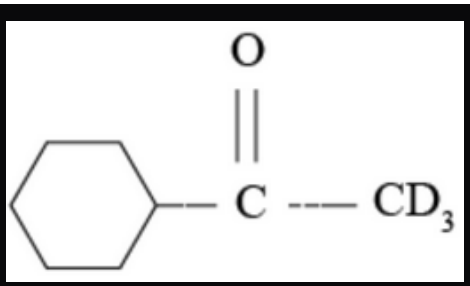
(D) All are correct

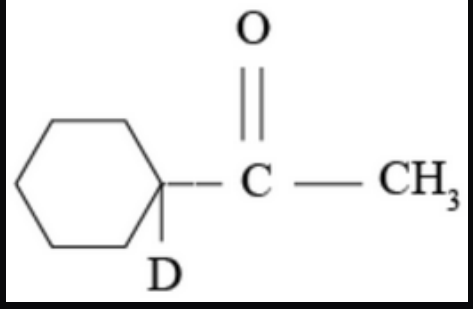
Q.32

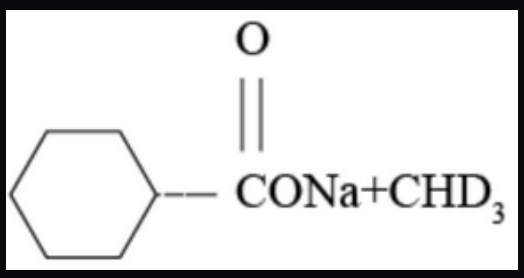
Final product (p) in the sequence of reaction is

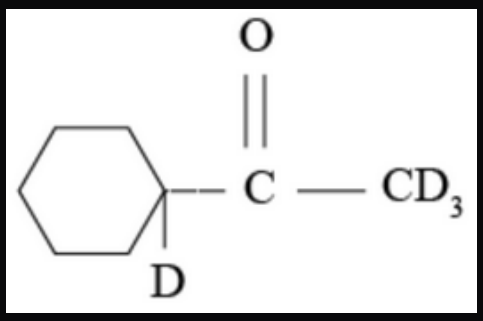


(A)



(B) 

(C) 

(D) 

Q.33

The threshold frequency of a metal is $1 \times 10^{15}\ s^{-1}$. The ratio of maximum kinetic energies of the photoelectrons when the metal is irradiated with radiations of frequencies $1.5 \times 10^{15}\ s^{-1}$ and $2 \times 10^{15}\ s^{-1}$ respectively would be

(A) $3 : 4$

(B) $1 : 2$

(C) $2 : 1$

(D) $4 : 3$

Q.34

Hydroxyl group is present in :

(A) Asparagine

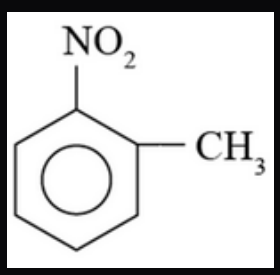
(B) Cysteine

(C) Serine

(D) Lysine

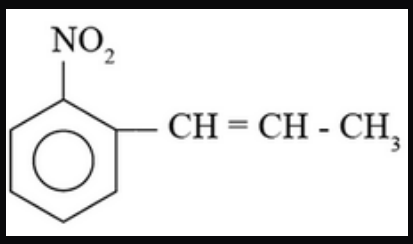
Q.35

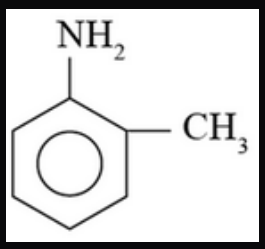
In the given reaction

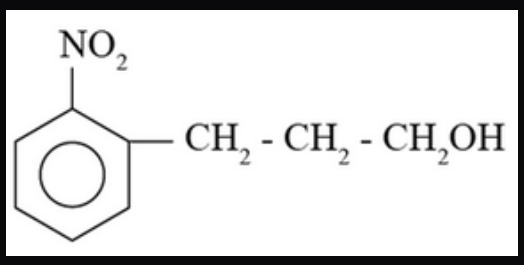


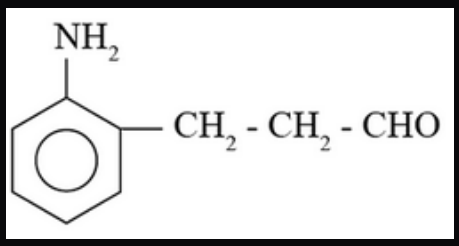
$+CH_3 - CHO \xrightarrow{\text{(i)NaOH (ii) } \Delta} [X]$

Then $[X]$ will be

(A) 

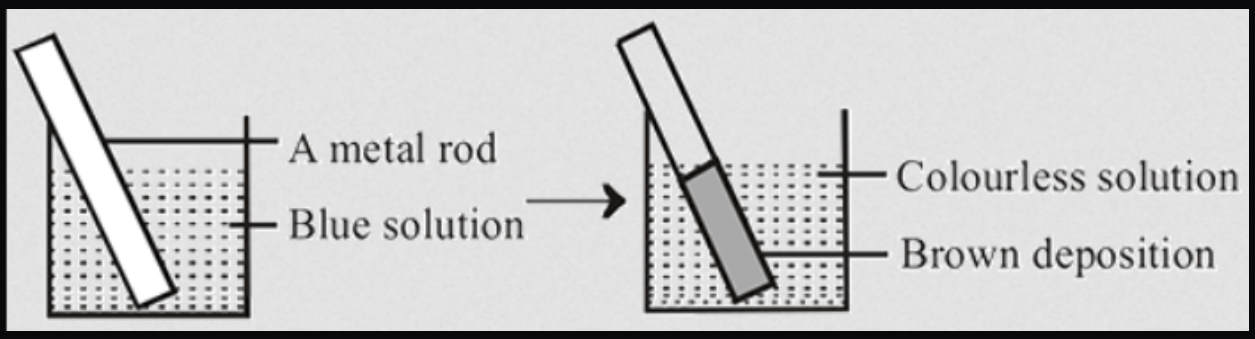
(B) 

(C) 

(D) 

Q.36

A redox reaction is shown in the diagrams. Identify the reaction.



(A) $Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

(B) $Cu(s) + 2Ag^{+}(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$

(C) $2Ag(s) + Cu^{2+}(aq) \longrightarrow 2Ag^{+}(aq) + Cu(s)$

(D) $Cu(s) + Zn^{2+}(aq) \longrightarrow Cu^{2+}(aq) + Zn(s)$

Q.37

A gaseous mixture of H_2 and CO_2 gas contains 33% by mass of CO_2. The vapour density of mixture is

(A) 2.5

(B) 1.5

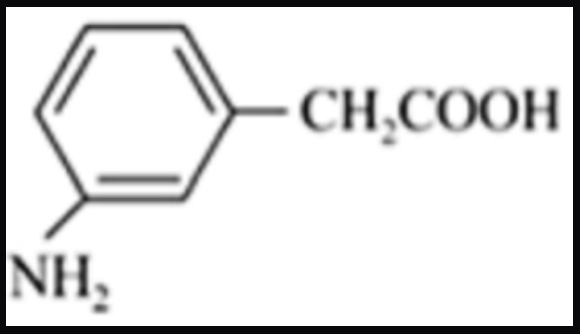
(C) 3.5

(D) 4.5

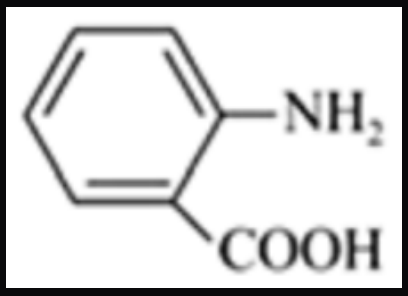
Q.38

Which of the following amino acids is converted to benzyne on diazotization followed by heating?

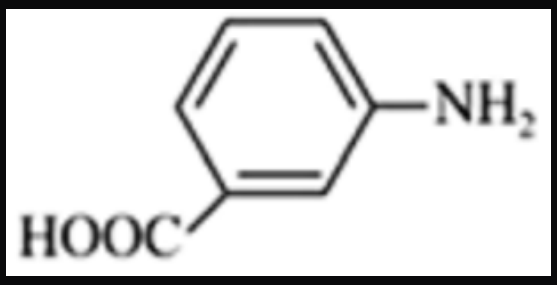
(A)



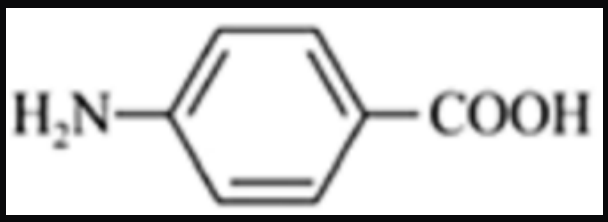
(B)



(C)



(D)



Q.39

Column I	Column II
A. He	i. Highest electron gain enthalpy
B. Cl	ii. Most electropositive element
C. Cs	iii. Strongest reducing agent
D. Li	iv. Highest ionisation energy

The correct match of the contents in column I with those in column II is

(A) A-(iii), B-(i), C-(ii), D-(iv)

(B) A-(iv), B-(iii), C-(ii), D-(i)

(C) A-(i), B-(ii), C-(iii), D-(iv)

(D) A-(iv), B-(i), C-(ii), D-(iii)

Q.40

The reagent (s) which can be used to distinguish acetophenone from benzophenone is (are)

(A) 2,4 - dinitrophenylhydrazine

(B) aqueous solution of $NaHSO_3$

(C) benedict reagent

(D) I_2 and Na_2CO_3.

Q.41

The sublimation energy of $I_2(s)$ is $57.3\ kJ\ mol^{-1}$ and the enthalpy of fusion is $15.5\ kJ\ mol^{-1}$. The enthalpy of vaporization of I_2 is

(A) $41.8\ kJ\ mol^{-1}$

(B) $-41.8\ kJ\ mol^{-1}$

(C) $72.8\ kJ\ mol^{-1}$

(D) $-72.8\ kJ\ mol^{-1}$

Q.42

When PCl_5 reacts with H_2SO_4, sulphuryl chloride (SO_2Cl_2) is formed. This shows that H_2SO_4:

(A) contains two hydroxyl groups in its molecule.

(B) contains two hydrogen (H) atoms in its molecule.

(C) contains one sulphur dioxide (SO_2) molecule.

(D) is a derivative of SO_3.

Q.43

Four statements regarding ethers are given below. Identify the FALSE statements.
(i) Ethers are named as alkoxy derivatives of alkane
(ii) The larger alkyl group along with oxygen atom is named as alkoxy group
(iii) The larger alkyl group is considered as the parent alkane
(iv) The position of the alkoxy group is indicated by the maximum number.

(A) (i) and (iii)

(B) (ii) and (iv)

(C) (ii) and (iii)

(D) (i) and (iv)

Q.44

In the Kjedahl's method for estimation of nitrogen present in soil sample, ammonia evolved from 0.75 g of sample neutralized $10ml$ of $1M$ H_2SO_4. The percentage of nitrogen in the soil is:

(A) 37.33

(B) 445.33

(C) 35.33

(D) 43.33

Q.45

Equivalent weight of $\mathrm{KMnO_4}$ when it is converted into $\mathrm{MnSO_4}$ is Where M = molar mass of $\mathrm{KMnO_4}$

(A) $\mathrm{M/5}$

(B) $\mathrm{M/6}$

(C) $\mathrm{M/3}$

(D) $\mathrm{M/2}$

Q.46

Find out total number of compound(s) in which at least half of $\mathrm{Cl^-}$ are ionizable

$\mathrm{CrCl_3.6NH_3, CrCl_3.5NH_3, CrCl_3.4NH_3, CrCl_3.3NH_3, PtCl_4.6NH_3, PtCl_4.5NH_3}$

$\mathrm{PtCl_4.4NH_3, PtCl_4.3NH_3, PtCl_4.2NH_3}$

Q.47

Half-life period of the radioactive element A is 10 days. Amount of A left on the end of 11th day starting with 1 mole A is $\left(\frac{1}{2}\right)^{\frac{n+6}{10}}$ mole. What is the value of n?

Q.48

How many grams of methyl alcohol should be added to $5\ \mathrm{L}$ tank of water to prevent its freezing at $268\ \mathrm{K}$? (kf for water is $1.86\ \mathrm{K\ kg\ mol^{-1}}$) (Consider solute to be non volatile)(mark answer to nearest whole number)

Q.49

The work done in an open vessel at $300\ \mathrm{K}$, when $112\ \mathrm{g}$ iron reacts with dilute HCl to give $\mathrm{FeCl_2}$, is nearly W, find 5W. Mark answer in calories.

Q.50

The specific conductance at $25^{\circ}\mathrm{C}$ of a saturated solution of AgCl in water is $1.826 \times 10^{-6}\ \mathrm{mho\ \ cm^{-1}}$. If $\Lambda_{\mathrm{AgCl}}{}^{\circ}$ is equal to $138.26\ \mathrm{mho\ \ cm^2\ \ mol^{-1}}$, find out the solubility of AgCl in water in milligrams per litre. (round off to the nearest integer)

Q.51

If the solution of the differential equation $y^3x^2\cos\left(x^3\right)dx+\sin\left(x^3\right)y^2dy=\frac{x}{3}dx$ is $2\sin\left(x^3\right)y^k=x^2+C$ (where C is an arbitrary constant), then the value of k is equal to

(A) 3

(B) 2

(C) 1

(D) 4

Q.52

Two distinct numbers x and y are chosen at random from the set $\{1,2,3\ldots.30\}$. The number of ways to select these two numbers such that x^2-y^2 is divisible by 3 are :

(A) 235

(B) 240

(C) 236

(D) 250

Q.53

The integral value of $\int_0^{\pi/4} [\sin x + [\cos x + [\tan x + [\sec x]]]]dx$ is equal to, where $[.]$ denotes the greatest integer function

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$

(D) π

Q.54

If A and B are two events defined on a sample space with the probabilities $P(A) = 0.5, P(B) = 0.69$ and $P\left(\frac{A}{B}\right) = 0.5$, then the value of $P\left(\frac{A}{A^c \cup B^c}\right)$ is equal to

(A) $\frac{2}{5}$

(B) $\frac{3}{13}$

(C) $\frac{31}{131}$

(D) $\frac{100}{131}$

Q.55

$\lim_{x \to 1^-} \frac{\sqrt{\pi} - \sqrt{2\sin^{-1} x}}{\sqrt{1-x}}$ is equal to

(A) $\sqrt{\pi}$

(B) $\sqrt{\frac{2}{\pi}}$

(C) $\frac{1}{\sqrt{2\pi}}$

(D) $\sqrt{\frac{\pi}{2}}$

Q.56

Let $\mathrm{Z} = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 1 & 2 \\ 3 & 1 & 0 \end{bmatrix}$ and $\mathrm{P} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$. If $\mathrm{Z} = \mathrm{PQ}^{-1}$, where Q is a square matrix of order 3, then the value of $\mathrm{Tr}((\mathrm{adjQ})\mathrm{P})$ is equal to (where $\mathrm{Tr}(\mathrm{A})$ represents the trace of a matrix A i.e. the sum of all the diagonal elements of the matrix A and adjB represents the adjoint matrix of matrix B)

(A) 3

(B) -1

(C) 4

(D) $\frac{6}{5}$

Q.57

The means of two samples of size 40 and 50 were found to be 54 and 63 respectively. Their standard deviations were 6 and 9 respectively. The variance of the combined sample of size 90 is

(A) 90

(B) 7

(C) 9

(D) 81

Q.58

Suppose a, b are integers and $a+b$ is a root of $x^2+ax+b=0$. What is the maximum possible values of b^2 ?

(A) 64

(B) 76

(C) 100

(D) None of these

Q.59

If the image of the point $M\left(\lambda, \lambda^2\right)$ on the line $x+y=\lambda^2$ is $N(0,2)$, then the sum of the square of all the possible values of λ is equal to

(A) 5

(B) 2

(C) 1

(D) 4

Q.60

The function $\mathrm{f(x)}=\sin^{-1}\left(2\mathrm{x}-\mathrm{x}^2\right)+\sqrt{2-\frac{1}{|\mathrm{x}|}}+\frac{1}{[\mathrm{x}^2]}$ is defined in the interval (where $[\cdot]$ is the greatest integer function)

(A) $\mathrm{x} \in \left(1-\sqrt{2}, 1\right)$

(B) $\mathrm{x} \in \left[1,1+\sqrt{2}\right]$

(C) $\mathrm{x} \in \left[1-\sqrt{2}, 1+\sqrt{2}\right]$

(D) $\mathrm{x} \in \left[1-\sqrt{2}, 2\right]$

Q.61

An ellipse of semi-axes $a,\ b$ slides between two perpendicular lines, then the locus of its foci is (the two lines being taken as the axes of coordinates)

(A) $\left(x^2+y^2\right)\left(x^2y^2+b^6\right)=4a^2x^2y^4$

(B) $\left(x^3+y^3\right)\left(x^3y^3+b^6\right)=4a^3x^3y^6$

(C) $\left(x^2+y^2\right)\left(x^2y^2+b^4\right)=4a^2x^2y^2$

(D) $\left(x^2-y^2\right)\left(x^2y^2-b^4\right)=4a^2x^2y^2$

Q.62

The number of solution(s) of the equation $z^2=4z+|z|^2+\frac{16}{|z|^3}$ is (where $z=x+iy,\ x,y\in R, i^2=-1$ and $x\neq 2$)

(A) 0

(B) 1

(C) 2

(D) 3

Q.63

In the binomial expansion of $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n$, the ratio of the 7th term from the beginning to the 7th term from the end is $1:6$, then the value of n is

(A) 13

(B) 16

(C) 9

(D) 23

Q.64

From a point P outside a circle with center at C, tangents PA and PB are drawn such that $\frac{1}{(CA)^2}+\frac{1}{(PA)^2}=\frac{1}{16}$, then the length of chord AB is

(A) 6

(B) 8

(C) 4

(D) 12

Q.65

The area inside the parabola $5x^2-y=0$ but outside the parabola $2x^2-y+9=0$ is equal to: (sq. unit)

(A) $6\sqrt{3}$

(B) $12\sqrt{3}$

(C) $3\sqrt{3}$

(D) $24\sqrt{3}$

Q.66

If $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$ and $\vec{b}=-3\hat{i}+\hat{j}-\hat{k}$ and $\vec{r}\times\vec{a}=\vec{b}\times\vec{a}, \vec{r}\times\vec{b}=\vec{a}\times\vec{b}$ then a unit vector in the direction of $\overrightarrow{\mathrm{r}}$ is

(A) $\frac{1}{3}(-2\hat{i}+\hat{j}-\hat{k})$

(B) $\frac{1}{3}(-2\hat{\mathbf{i}}-\hat{\mathbf{j}}+2\hat{\mathbf{k}})$

(C) $\frac{1}{3}(-2\hat{i}-\hat{j}-2\hat{k})$

(D) None of these

Q.67

The digits $A,\ B$ and C are such that the three digit numbers $A88,\ 6B8,\ 86C$ are divisible by 72, then the determinant $\begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ 8 & 8 & C \end{vmatrix}$ is divisible by

(A) 72

(B) 144

(C) 288

(D) All of these

Q.68

If $\tan\theta = 3\tan\phi$, then the maximum value of $\tan^2(\theta - \phi)$ is (where, $\tan\phi > 0$)

(A) 1

(B) $\frac{1}{3}$

(C) 2

(D) 4

Q.69

Let $\mathrm{f(x)}$ be a second degree polynomial function such that $\ell\mathrm{n}(\ \mathrm{f(x)}) \ > \ 0 \ \forall \ \mathrm{x} \ \in \ \mathrm{R}$ and the equation $\mathrm{f}\ \prime\prime(\mathrm{x}) + \ 21\ \mathrm{f(x)} = \ 0$ has no real roots. If $\mathrm{g(x)} = \ \mathrm{e}^{21\mathrm{x}}\ \mathrm{f(x)}$. Then

(A) $\mathrm{g(x)}$ is an increasing function

(B) $\mathrm{g(x)}$ is an even function

(C) $\mathrm{g(x)}$ is a decreasing function

(D) the graph of $\mathrm{g(x)}$ cuts x -axis exactly once

Q.70

Let the equation of a line is $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{3}$. An insect starts flying from $P(1,3,2)$ in a straight line meeting the given line at a point $R(a,b,c)$ and then goes to the point $Q(6,7,5)$ in a straight line such that PR is perpendicular to RQ. Then, the least value of $7(a+b+c)$ is equal to

(A) 105

(B) 45

(C) 10

(D) 7

Q.71

If $I = \int_0^{16} \frac{x^{\frac{1}{4}}}{1+\sqrt{x}}dx = k + 4\tan^{-1} m$, then $3k - m$ is equal to

Q.72

Let $f : R \to R$ be a polynomial function satisfying the equation $f(f(x) - 2y) = 2x - 3y + f(f(y) - x), \forall x, y \in R$, then the value of $f(9) - f(3)$ is equal to

Q.73

Find the angle in degree between two lines whose direction cosines are given by $\ell+\mathrm{m}+n=0, \ell^2+\mathrm{m}^2-n^2=0$

Q.74

If the total number of positive integral solutions of

$15<x_1+x_2+x_3 \leq 20$ is k, then the value of k is equal to

Q.75

If $A_1, A_2, A_3 \ldots \ldots . A_{20}$ are 20 skew-symmetric matrices of same order and $B=\sum_{r=1}^{20} 2 r\left(A_r\right)^{(2 r+1)}$, then the sum of the principal diagonal elements of matrix B is equal to

Answers & Solutions

Q.1 Answer:

$\frac{\mathrm{n}}{2}$

Solution:

$$f=\frac{1}{2 L} \sqrt{\frac{T}{\mu}}=\frac{1}{2 L} \sqrt{\frac{T}{\rho \cdot \pi r^2}}$$

$r \rightarrow 2$ times

$f \rightarrow \frac{1}{2}$ times

No. of oscillation ($\sec \rightarrow$ half)

Q.2 Answer:

$\mathrm{a} \rightarrow \mathrm{r}, \mathrm{s}$, $\mathrm{b} \rightarrow \mathrm{p}, \mathrm{s}$, $\mathrm{c} \rightarrow \mathrm{r}, \mathrm{s}$, $\mathrm{d} \rightarrow \mathrm{q}$

Solution:

$(\mathrm{a} \rightarrow \mathrm{r})$

Enthalpy is a state function and this is a function of internal energy, pressure and volume.

$(\mathrm{b} \rightarrow \mathrm{p})$

Temperature is independent of bulk matter of the system.

Hence, it is an intrinsic property but it is a state variable.

$(\mathrm{c} \rightarrow \mathrm{r})$

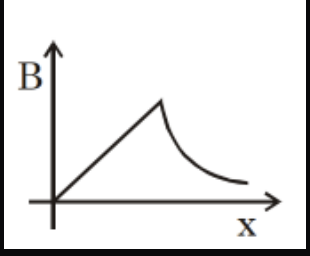
Free energy is a state function as it depends on the state of the system as well as H, T and S.

It is also a function of U, P and V.

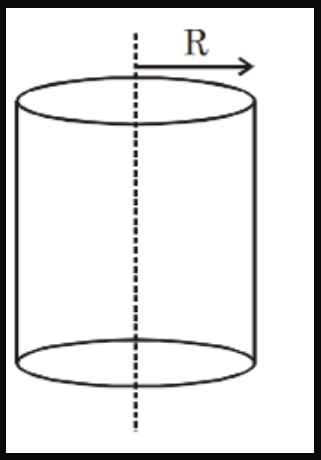
$(d \rightarrow q)$

Work is a path function as the amount of work done depends upon the path undertaken.

Q.3 Answer:



Solution:



Applying Ampere's Law

For inside

$$B2\pi r = \mu_0 \, \mathrm{J}\pi r^2$$

$$\Rightarrow B \propto r$$

For outside

$$\mathrm{B}2\pi \mathrm{r} = \mu_0 \, \mathrm{J}\pi \mathrm{R}^2$$
$$\Rightarrow \mathrm{B} \propto \frac{1}{\mathrm{r}}$$

Q.4 Answer:

30 ms^{-1}

Solution:

Given the radius of the circular path $\mathrm{r} = 150 \, \mathrm{m}$.

The coefficient of friction $\mu = 0.6$.

The maximum velocity for driving without slipping is given by,

$\mathrm{V} = \sqrt{\mu \mathrm{rg}}$.

Substituting the given values, we have,

$\mathrm{v} = \sqrt{0.6 \times 150 \times 10} = \sqrt{900} = \ 30 \ \mathrm{ms}^{-1}$

Q.5 Answer:

$4\mathrm{R}$

Solution:

As, $T^2 \propto R^3$

$\therefore \frac{\mathrm{T_A}^2}{\mathrm{T_B}^2} = \frac{\mathrm{R_A}^3}{\mathrm{R_B}^3}$

$\left(\frac{\mathrm{T_A}}{8\mathrm{T_A}}\right)^2 = \left(\frac{\mathrm{R_A}}{\mathrm{R_B}}\right)^3 \quad [\because \mathrm{T_B} = 8\mathrm{T_A}]$

$\left(\frac{1}{8}\right)^2 = \left(\frac{R_A}{R_B}\right)^3$

$\Rightarrow \frac{1}{64} = \left(\frac{\mathrm{R_A}}{\mathrm{R_B}}\right)^3$

$\Rightarrow \left(\frac{1}{4}\right)^3 = \left(\frac{\mathrm{R_A}}{\mathrm{R_B}}\right)^3$

$\Rightarrow \frac{1}{4} = \frac{\mathrm{R_A}}{\mathrm{R_B}}$

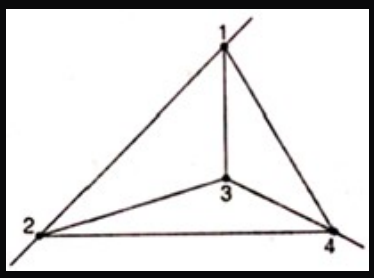
$\Rightarrow \mathrm{R_B} = 4\mathrm{R_A} = 4\mathrm{R}$

Q.6 Answer:

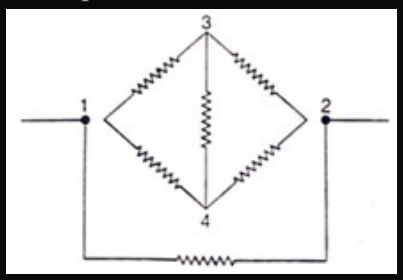
$\frac{r}{2}$

Solution:

Six wires each of resistance r form a tetrahedron as shown in the following figure.



The equivalent circuit of this tetrahedron will be,



It is circuit of Wheatstone bridge. The equivalent resistance of upper circuit is,

$\frac{1}{R} = \frac{1}{2r} + \frac{1}{2r} = \frac{2}{2r} = \frac{1}{r}$

It will in parallel with outer resistance,

$\frac{1}{R_{eq}} = \frac{1}{r} + \frac{1}{r} = \frac{2}{r}$

$R_{eq} = \frac{r}{2}$

Q.7 Answer:

$I_2 = 9.7\ \text{A}$

Solution:

$Ge-$**diode will start conducting at** $0.3\ \text{V}$ **and** $Si-$**diode will not conduct at** $0.3\ \text{V}$.

Voltage change $\text{V}_0 = 10 - 0.3 = 9.7\ \text{V}$

Current I **will be same as** $I_2 = V_0/R = 9.7/1 = 9.7\ \text{A}$

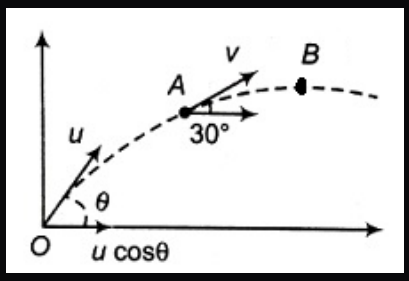
Q.8 Answer:

$20\sqrt{3}\ \text{m s}^{-1},\ \ 60^\circ$

Solution:

Let in 2s body reaches upto point A and after one more second upto point B.

Total time of ascent for a body is given as 3s



i.e., $\frac{u \sin\theta}{g} = 3$

$\therefore u \sin\theta = 10 \times 3 \ = 30$ **.....(i)**

Horizontal component of velocity remains always constant

$u \cos\theta = v \cos 30^\circ$ **.....(ii)**

For vertical upward motion between point O and A

$v \sin 30^o = u \sin\theta - g \times 2$ **[Using v = u - gt]**

$v \sin 30^o = 30 - 20\ [As\ u \sin\theta = 30]$

$\therefore v = 20\ \text{m s}^{-1}$

Substituting this value v **in equation (ii), we get**

$u \cos\theta = 20 \cos 30^\circ = 10\sqrt{3}$ **.....(iii)**

From equations (i) and (ii)

$u = 20\sqrt{3}\ \mathrm{m\ s^{-1}}$ **and** $\theta = 60^\circ$

Q.9 Answer:

Assertion is true, reason is true, reason is correct explanation for assertion.

Solution:

If, $R_\mathrm{A} < R_B$**,**

Now recall the formula of excess pressure inside the liquid drop,

$p_\mathrm{A} = \frac{4T}{R_\mathrm{A}}$ & $p_\mathrm{B} = \frac{4T}{R_\mathrm{B}}$**,**

$\Rightarrow p_\mathrm{A} > p_\mathrm{B}$**, and we know that gases passes from high pressure to low pressure it means bubble** A **will become small and** B **will become larger.**

Q.10 Answer:

None of these.

Solution:

Velocity of electromagnetic wave

$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8\ \mathrm{m\ s^{-1}}$

It is independent of amplitude, frequency and wavelength of electromagnetic wave.

Q.11 Answer:

$\frac{100}{3}\ \mathrm{m\ s^{-1}}$

Solution:

Current flowing in circuit is $I = \ 2\ \mathrm{A}$ **and total resistance** $R = 8 + 4 = 12\ \Omega$

$I = \frac{\text{Potential difference}}{\text{Resistance}}$

$\therefore$**Potential difference is** $\varepsilon = 2 \times 12 = 24\ \mathrm{V}$

The induced emf in the loop is given by,

$\varepsilon = BLv$**; where** B **is magnetic field and** v **is velocity of the loop.**

Here, $L = AD \sin 37^\circ = 0.3 \times \frac{3}{5} = 0.18\ \mathrm{m}$**.**

Velocity will be given by,

$v = \frac{24}{Bl} = \frac{24}{4 \times 1.08} v = \frac{100}{3}\ \mathrm{m\ s^{-1}}$

Q.12 Answer:

If Assertion is false but Reason is true.

Solution:

Frequency of an EM wave is the property of source. It does not depend on number of photons emitted. Hence, Assertion is false and Reason is true.

Q.13 Answer:

$\sqrt{\frac{3}{2}}$

Solution:

From the question, we know that the second angle is critical, and that the sum of the two angles inside the slab is 90°.

Applying Snell's law for first surface,

$1 \times \sin\ 45^\circ = \mu \times \sin\ r$

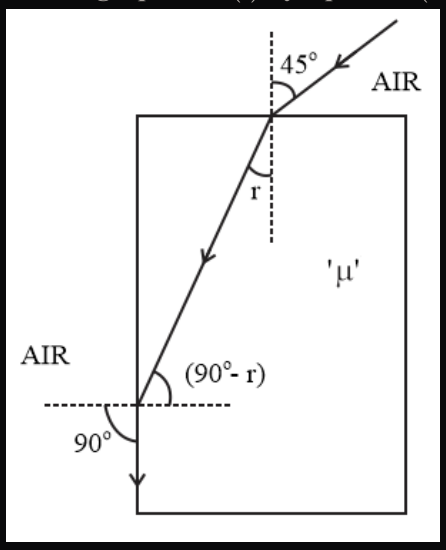
$\mu \times \sin\ r = \frac{1}{\sqrt{2}} \ldots (\mathrm{i})$

Critical angle condition for the second surface,

$\mu \times \sin(90 - r)\ = 1 \times \sin\ 90^\circ$

$\Rightarrow \mu \times \cos\ r = 1 \ldots (\mathrm{ii})$

Dividing equation (i) by equation (ii), we get :



$\Rightarrow \tan\ r = \frac{1}{\sqrt{2}} \Rightarrow \sin\ r = \frac{1}{\sqrt{3}}.$

Put this value in equation (i),

$\Rightarrow \mu = \sqrt{\frac{3}{2}}.$

Q.14 Answer:

Either dark or bright depending on the position of S

Solution:

Path difference, $x = (SS_1 + S_1O) - (SS_2 + S_2O).$

If, $x = n\lambda$, then central fringe at O will be bright.

If, $x = (2n - 1)\lambda/2$, the central fringe at O will be dark.

Q.15 Answer:

Time

Solution:

Dimensions of $E = \left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right]$

Dimensions of $G = \left[\mathrm{M}^{-1}\ \mathrm{L}^3\ \mathrm{T}^{-2}\right]$

Dimensions of $I = \left[\mathrm{MLT}^{-1}\right]$

And dimension of $M = [\mathrm{M}]$

$\therefore$ **Dimensions of** $\frac{GIM^2}{E^2} = \frac{\left[\mathrm{M}^{-1}\ \mathrm{L}^3\ \mathrm{T}^{-2}\right]\left[\mathrm{MLT}^{-1}\right]\left[\mathrm{M}^2\right]}{\left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right]^2}$

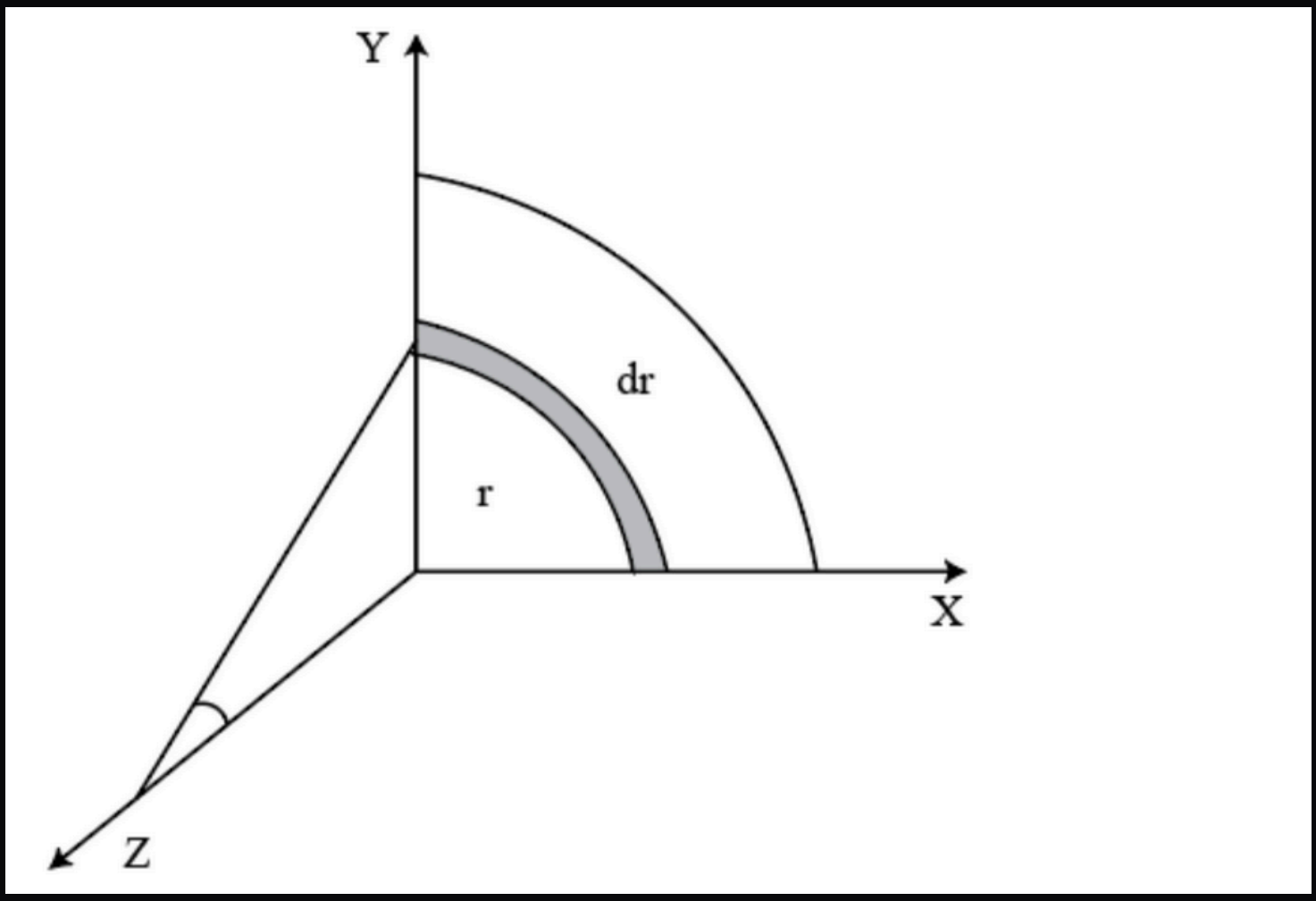
$= [\mathrm{T}]$

$=$ **Dimensions of time**

Q.16 Answer:

$\frac{\sigma}{8\epsilon_0}\left[\sqrt{R^2+Z^2}-Z\right]$

Solution:



$V = \int_0^R \frac{k\alpha\pi r dr}{\alpha\sqrt{r^2+z^2}} = \frac{\sigma}{8\epsilon_0}\int_0^R \frac{r dr}{\sqrt{r^2+z^2}}$

$t^2 = r^2 + z^2$

$tdt = rdr$

$V = \frac{\sigma}{8\epsilon_0} \int_z^{\sqrt{2^2+R^2}} dt = \frac{\sigma}{8\epsilon_0} \left(\sqrt{R^2+z^2} - z\right)$

Q.17 Answer:

$-5J$

Solution:

$W_{AB} + W_{\mathrm{BC}} + W_{CA} = 5$ **(change in internal energy in a cyclic process equal to zero.)**

$W_{CA} = 5 - (W_{AB} + W_{BC}) = 5 - (10 \times 1) = -5J$

Q.18 Answer:

$104\mathrm{MeV}$

Solution:

The binding energy of nucleus may be defined as the energy equivalent to the mass defect of the nucleus.

If Δm is mass defect than according to Einstein's mass energy relation.

Binding Energy

$= \Delta mc^2 = [\{Zm_p + (A-Z)m_n] - M]\, c^2$

$= (7 \times 1.00783 + 7 \times 1.00867 - 14.00307)c^2$

or $\mathrm{BE} = 0.1124 \times 931.5\mathrm{MeV}$

or $\mathrm{BE} = 104.6\mathrm{MeV}$

Q.19 Answer:

$0.98\ \mathrm{m/s^2}$

Solution:

Limiting friction between block and slab

$= \mu_s m_A g = 0.6 \times 10 \times 9.8 = 58.8\ \mathrm{N}$

But applied force on block A is $100\ \mathrm{N}$. So the block will slip over a slab.

Now kinetic friction works between block and slab

$\mathrm{F_k} = \mu_\mathrm{k} \mathrm{m_A} \mathrm{g} = 0.4 \times 10 \times 9.8 = 39.2\ \mathrm{N}$

This kinetic friction helps to move the slab

$\therefore$ Accleration of slab $= \frac{39.2}{\mathrm{m_B}} = \frac{39.2}{40} = 0.98\ \mathrm{m/s^2}$

Q.20 Answer:

$182 : 1$

Solution:

Magnetic force is Bqv which is always perpendicular to the velocity of charged particle, so particle executes circular motion.

So, centripetal force is provided by magnetic force.

Hence, for proton, $\frac{m_p v_p^2}{r_p} = Bqv_p$

and for electron, $\frac{m_\theta v_\theta^2}{r_e} = Bqv_e$

$$\Rightarrow r_p = \frac{m_p v_p}{Bq}$$
$$\text{and } r_\theta = \frac{m_\theta V_\theta}{Bq}$$
$$\therefore \frac{r_p}{\theta} = \frac{m_p}{m_\theta} \cdot \frac{v_p}{v_\theta}$$
$$= \frac{1820 m_\theta}{m_\theta} \times \frac{v}{10v}$$

(given, $m_p = 1820 m_c$ and

$v_p = v, v_\theta = 10\mathrm{v}$)
$= 182 : 1$

Q.21 Answer:

—

Solution:

As we know that

$$\int \vec{E} \cdot \overrightarrow{dl} = -\frac{d\phi}{dt}$$

$$\Rightarrow \quad E \times 2\pi R = \left(\pi R^2\right) \frac{dB}{dt}$$

$$\Rightarrow \quad E = \frac{R}{2} \times 8 = 2N/C$$

$$\Rightarrow \quad F = qE = 16N, T = 10s$$

$$\Rightarrow \quad FR = mR^2 \times \alpha \Rightarrow \alpha = \frac{F}{mR} = \frac{16}{0.1 \times 0.5} = \frac{1600}{5} = 320$$

$$\omega = \omega_0 + \alpha t = 320 \times 10$$

$$K.E. = \frac{1}{2} I\omega^2 = \frac{1}{2} \times 0.1 \times \frac{1}{4} \times (3200)^2$$

Q.22 Answer:

—

Solution:

When the ball comes back into the hands of the boy, the displacement of boy and ball will be same.

Misplaced \hline

$\because$ & $S_{\text{Boy}} = \text{S}_{\text{Ball}}$

$10t + \frac{1}{2} \times 2 \times t^2 = -10t + \frac{1}{2} \times 10 \times t^2$

$20t = 5t^2 - t^2,$

$20t = 4t^2$

$t = 5 \text{ s}$

Q.23 Answer:

—

Solution:

Mass , $\text{m} = 0.2 \text{ kg}$

Total height, $\text{h} = 0.2 + 2 = 2.2 \text{ m}$

Work done = Difference in potential energy.

F.S = mgh where S **is the distance for ehich the force is applied by hand,** $\text{S} = 0.2 \text{ m}$

$$\text{F} = \frac{\text{mgh}}{\text{S}} = \frac{0.2 \times 10 \times 2.2}{0.2}$$
$$\text{F} = 22 \text{ N}$$

Q.24 Answer:

—

Solution:

Here, $a = 1 \text{ cm} = 0.01 \text{ m}$**. The mass will remain in contact with surface,**

if $mg = m\omega^2 a$

or $\omega = \sqrt{\frac{g}{a}}$

or $2\pi v = \sqrt{\frac{g}{a}}$

Or

$\text{v} = \frac{1}{2\pi}\sqrt{\frac{g}{a}}$

$$= \frac{7}{2 \times 22}\sqrt{\frac{980}{1}}$$
$$= 4.9 \text{ Hz} \approx 5 \text{ Hz}$$

$= 4.9 \text{ Hz} \approx 5 \text{ Hz}$

Q.25 Answer:

—

Solution:

Given that,
The mass of water is $m = 5 \text{ kg}$**.**
The change in temperature is $\triangle T = 100°\text{C} - 20°\text{C} = 80°\text{C}$**.**
The specific heat of water is $c = 1 \text{ kcal kg}^{-1} \text{ °C}^{-1}$**.**
The heat energy required is $Q = mc \triangle T = 5 \times 1 \times 80 = 400 \text{ kcal}$**.**

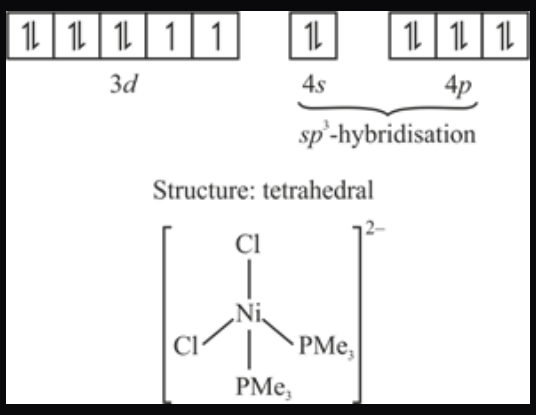
Q.26 Answer:

Zero

Solution:

$[\text{NiCl}_2(\text{PMe}_3)_2]$

Ni^{2+} **in** $[\text{NiCl}_2(\text{PMe}_3)_2]$



So, no isomer is possible, so answer is zero.

Q.27 Answer:

3 > 2 > 4 > 1 > 5

Solution:

The alcohols (3) and (2) are both 3°, but alcohol (3) gives a more substituted alkene. Alcohol (4) and (1) are both 2°, but alcohol (4) can give a more substituted alkene and alcohol (5) is 1°. Rate of dehydration of alcohols with concentration H_2SO_4 follows the order $3 > 2 > 4 > 1 > 5$

Q.28 Answer:

CrO_5

Solution:

The blue colour is due to the formation of CrO_5

$$CrO_4^{2-} + 2H^+ + 2H_2O_2 \rightarrow \underset{\text{(Blue)}}{CrO_5} + 3H_2O$$

Q.29 Answer:

The second reaction is more sensitive towards temperature variation

Solution:

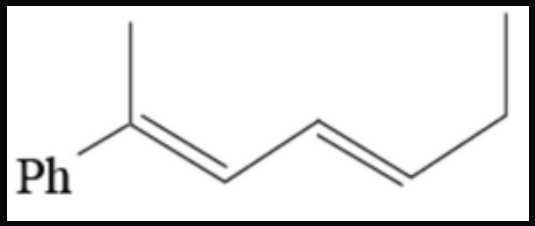
Arrhenius equation; $d(\ln k) = \frac{E_a}{RT^2} dT$

Here, T **and** dT **same and positive** $d(\ln k)\ \alpha\ E_a$

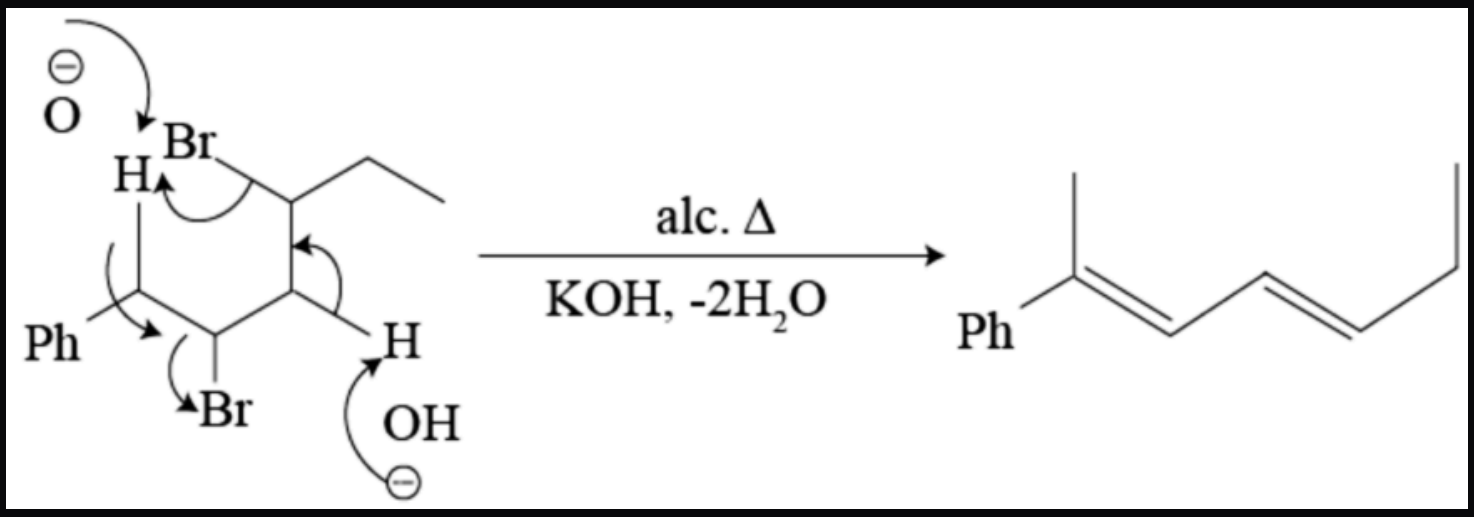
$$\ln k_2 - \ln k_1\ \alpha\ E_a$$
$$\ln \frac{K_2}{K_1}\ \alpha\ E_a$$

So, if temperature increases, then rate of reaction having more E_a increases more sharply.

Q.30 Answer:



Solution:



This product is highly stable conjugate di-ene system with benzene ring also so it is formed here as major product.

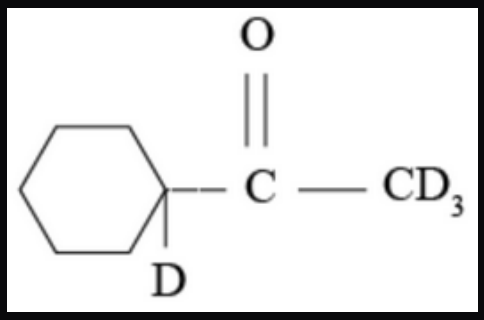
Q.31 Answer:

(i) and (ii)

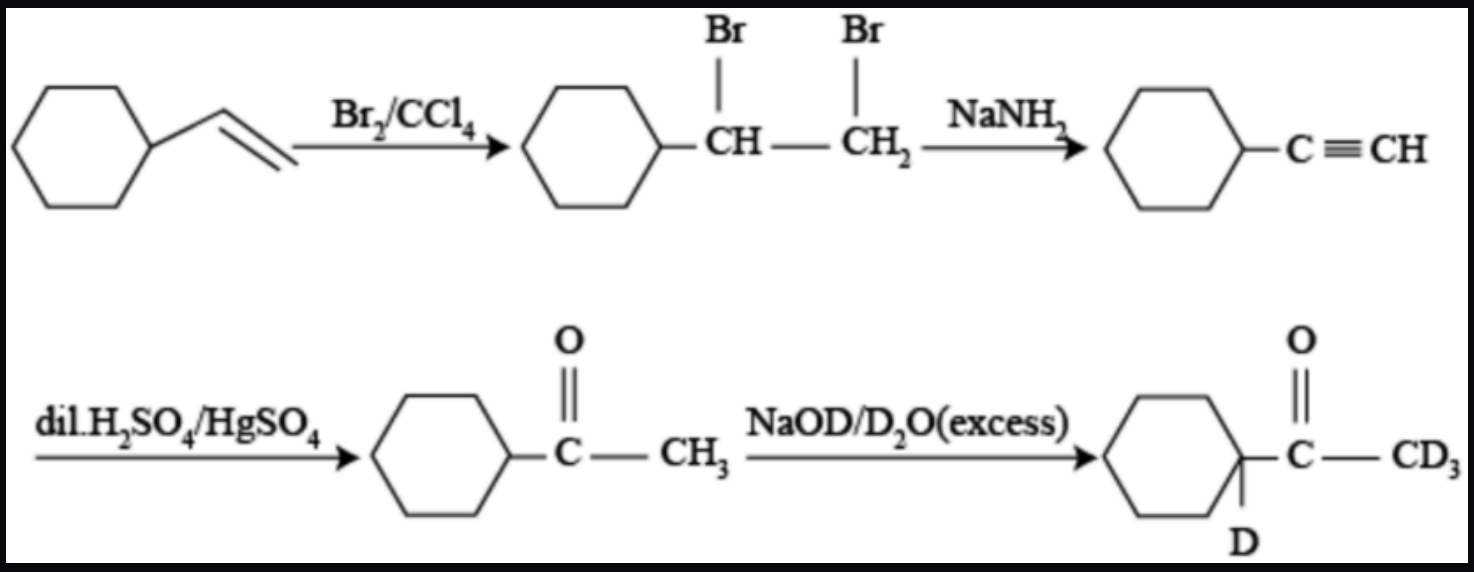
Solution:

NO : $\sigma 1s^2\sigma^* 1s^2\sigma 2s^2\sigma^* 2s^2\sigma 2p_z^2\pi 2p_x^2 = \pi 2p_y^2\pi^* 2p_x^1 = \pi^* 2p_y^0$ **and it is similar to** O_2^+ .

Q.32 Answer:



Solution:



Q.33 Answer:

1 : 2

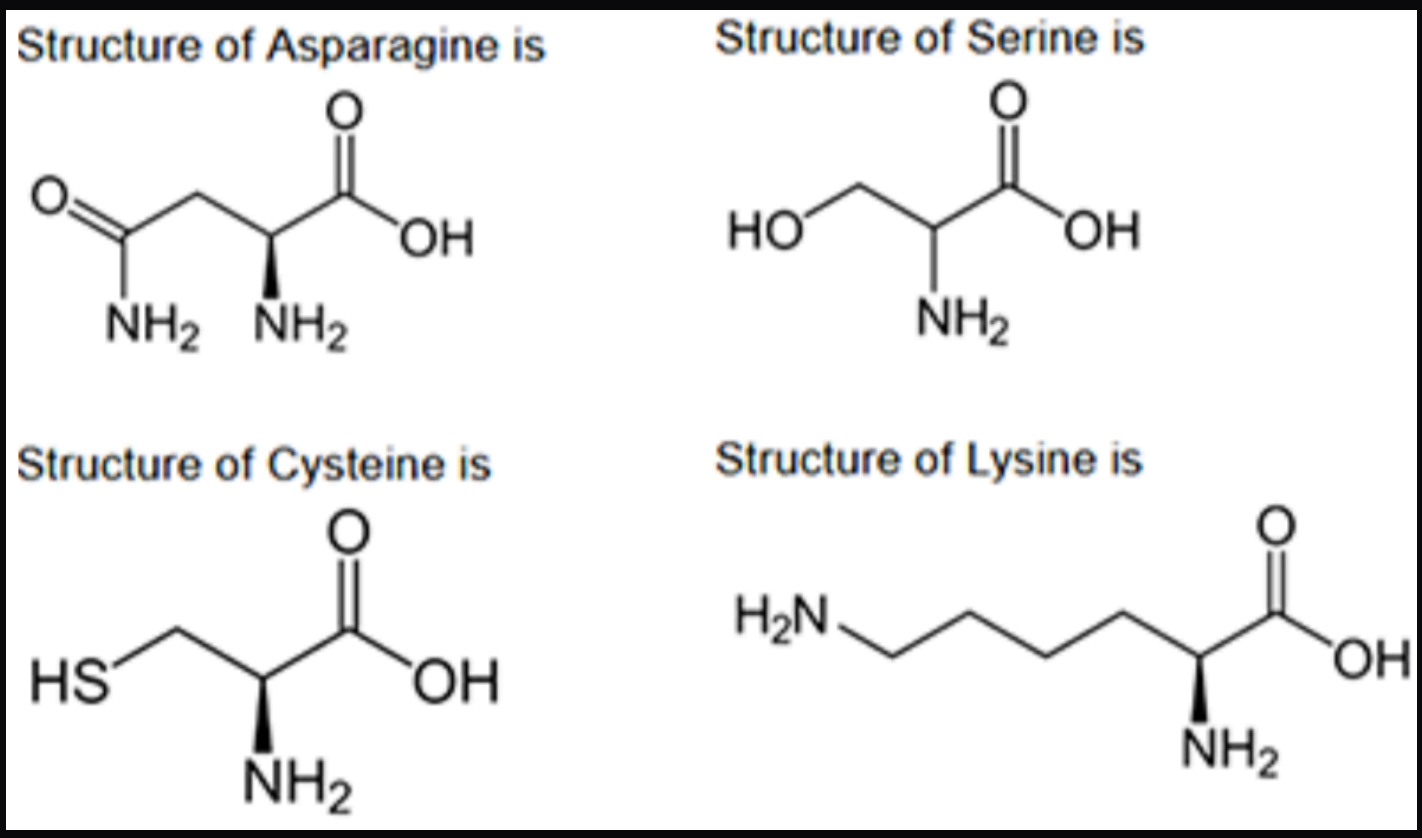
Solution:

$$\frac{(K.E.)_1}{(K.E.)_2} = \frac{v_1 - v_0}{v_2 - v_0} = \frac{\left(1.5 \times 10^{15} - 1 \times 10^{15}\right)}{(2.0 \times 10^{15} - 1 \times 10^{15})} = \frac{1}{2}$$

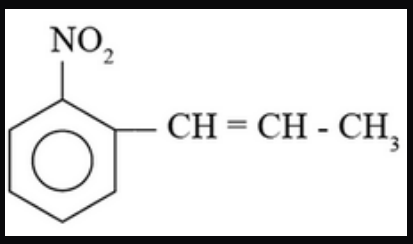
Q.34 Answer:

Serine

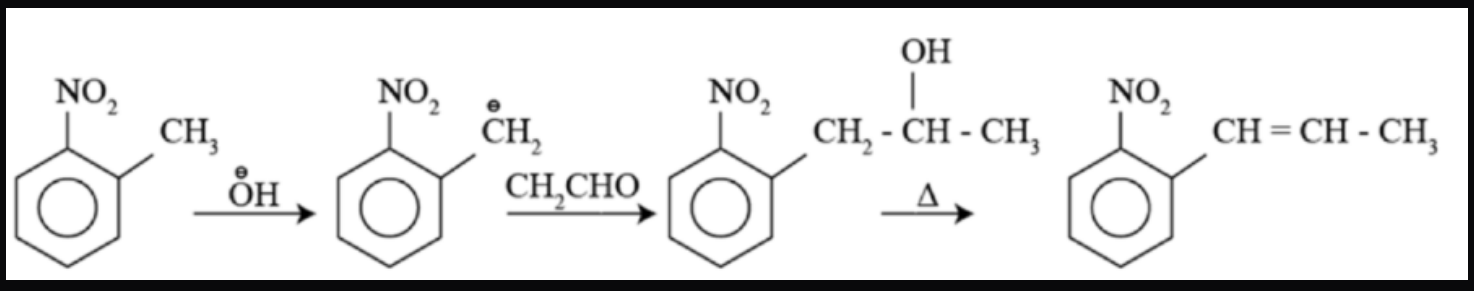
Solution:



Q.35 Answer:



Solution:



Q.36 Answer:

$\mathrm{Zn}(s) + \mathrm{Cu}^{2+}(aq) \longrightarrow \mathrm{Zn}^{2+}(aq) + \mathrm{Cu}(s)$

Solution:

Zinc rod dipped in blue copper sulphate solution is oxidised to Zn^{2+} and Cu^{2+} are reduced to Cu and get deposited on zinc rod.

Q.37 Answer:

1.5

Solution:

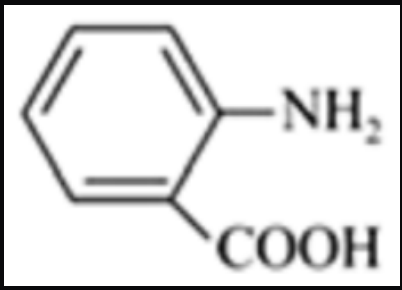
Number of moles of CO_2 in 100 g mixture $= \frac{33}{44} = 0.75$

Number of moles of H_2 in 100 g mixture $= \frac{67}{2} = 33.5$

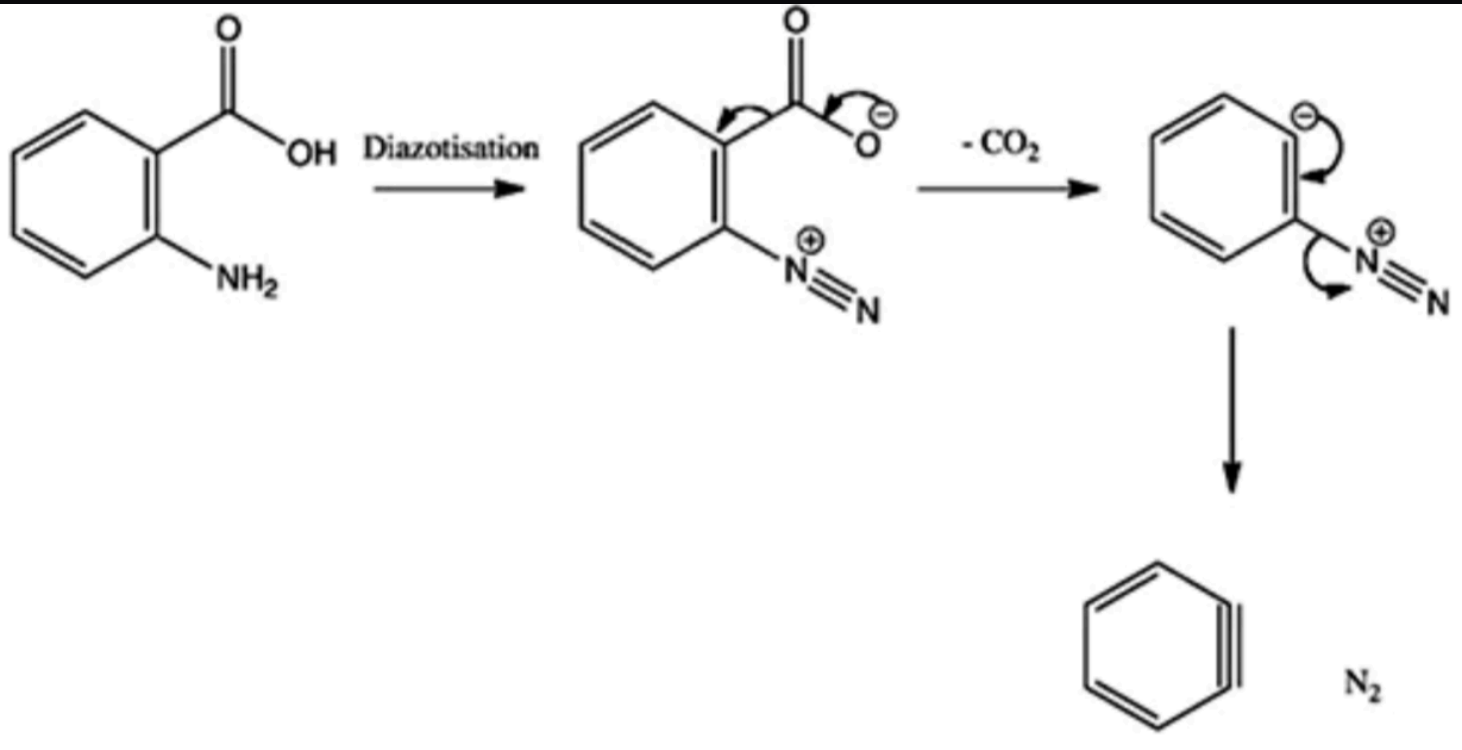
$M_{av} = \frac{100}{33.5} = 2.98 \approx 3$

Vapour density $= \frac{3}{2} = 1.5$

Q.38 Answer:



Solution:



Q.39 Answer:

A-(iv), B-(i), C-(ii), D-(iii)

Solution:

He ⇒ **Inert gas**

Cl ⇒ **Electron gain enthalpy is highest**

Cs ⇒ **Most electropositive metal**

Li ⇒ **Strong reducing agent**

He **has the electronic configuration of** $1s^2$**. It is completely filled electronic configuration, due to this, it has the highest ionisation enthalpy.**

$Cl \longrightarrow Cl^-$ **gives the highest** $\Delta_{eg}H$ **as** Cl^- **after getting one electron, it will achieve the stable electronic configuration** $18[Ar]$**. Due to this, large amount of energy is released during this process.**

$_{20}Cs \xrightarrow[-2e^-]{} Cs^{2+}$

Alkaline earth metal and present in the fourth period, the outermost electron is far away from the nucleus. As a result, less energy is required to remove the electron and easily donate the electron, hence, shows maximum metallic nature.

E^{o}_{red} of Li is –3. 07 V, a strong reducing agent.

It has very high hydration energy due to smaller in size. The hydration energy is directly proportional to the charge density of the ion.

Q.40 Answer:

I_2 and Na_2CO_3.

Solution:

I_2 and Na_2CO_3 react with acetophenone $(C_6H_5COCH_3)$ to give yellow ppt. of CHI_3 but benzophenone $(C_6H_5COC_6H_5)$ does not and hence can be used to distinguish between them.

Q.41 Answer:

$41.8\ kJ\ mol^{-1}$

Solution:

The sublimation is the process in which the solid converted into its vapour state directly. The sublimation, fusion and vaporization are related as

$\Delta H_{sub} = \Delta H_{fusion} + \Delta H_{vap}. \Delta H_{vap} = \Delta H_{sub} - \Delta H_{fusion}. \Delta H_{vap} = 57.3\ kJ\ mol^{-1} – 15.5\, kJ\ mol^{-1}. \Delta H_{vap} = 41.8\ kJ\ mol^{-1}.$

Q.42 Answer:

contains two hydroxyl groups in its molecule.

Solution:

When PCl_5 reacts with sulphuric acid, sulphuryl chloride $(SO_2\,Cl_2)$ is formed as the final product. It shows that it has two hydroxyl groups in its structure. This shows that $H_2\,SO_4$ contain two hydroxyl groups in its molecule. The reaction for this:

$HO - SO - OH + PCl_5 \rightarrow Cl - SO_2 - Cl + POCl_3 + 2\,HCl$

Q.43 Answer:

(ii) and (iv)

Solution:

Ethers are compounds having two alkyl or aryl groups bonded to an oxygen atom, as in the formula $R_1 - O - R_2$ The ether functional group does not have a characteristic IUPAC nomenclature suffix, so it is necessary to designate it as a substituent. To do so the common alkoxy substituents are given names derived from their alkyl component. The smaller, shorter alkyl group becomes the alkoxy substituent. The larger, longer alkyl group side becomes the alkane base name.The numbering priority is given to the carbon closest to the oxygen.

For example, $CH_3\,CH_2 - O - CH_3$ is 1-methoxyethane.

Q.44 Answer:

37.33

Solution:

10ml of $1\text{M}\,H_2SO_4$ corresponds to 10mmol

It will neutralize 20mmol of NH_3.

Weight of $\text{N} = \frac{14\times20}{1000}\ \text{g} = 0.280\ \text{g}$

Thus $0.75\ \text{g}$ of sample of organic compound contains $0.280\ \text{g}$ of nitrogen.

The percentage of nitrogen in the organic sample is

$\frac{0.280}{0.75} \times 100 = 37.33\%.$

Q.45 Answer:

$\text{M}/5$

Solution:

$MnO_4^- + 5e^- \rightarrow Mn^{2+}$

So equivalent weight of $KMnO_4$

$= \frac{\text{Molar mass of } KMnO_4(\text{M})}{\text{No. of mol of electrons gained by 1 mol of } KMnO_4} = \frac{\text{M}}{5}$

Q.46 Answer:

—

Solution:

Total number of compound(s) in which at least half of Cl^- are ionizable

$CrCl_3.6NH_3, CrCl_3.5NH_3, PtCl_4.6NH_3, PtCl_4.5NH_3, PtCl_4.4NH_3$

Q.47 Answer:

—

Solution:

$\text{A}_t = \text{A}_0 \times \left(\frac{1}{2}\right)^{\text{number of half life}}$

$\left(\frac{1}{2}\right)^{\left(\frac{n+6}{10}\right)} = \left(\frac{1}{2}\right)^{\left(\frac{11}{10}\right)}$

$n = 5$

Q.48 Answer:

—

Solution:

$$\Delta T_f = (273 - 268\ \text{K}) = 5\ \text{K}$$
$$\Delta T_f = K_f m \Rightarrow 5 = 1.86 \times \frac{n}{5}$$
$$\Rightarrow n = 13.44 \text{ moles of Methyl alcohol}$$
$$\Rightarrow m = (13.44 \times 32)\text{g} \simeq 430\ \text{g}$$

Q.49 Answer:

—

Solution:

$\mathrm{Fe} + 2\mathrm{HCl} \rightarrow \mathrm{FeCl_2} + \mathrm{H_2}$

Moles of $Fe = \frac{112}{56} = 2$

Moles of H_2 **formed** $= 2$ **Work done** $= P\left(V_2 - V_1\right) = P.V_{H_2}$ $[V_2 = V_{H_2}$ **and** $V_1 = 0$ **(for solid+liquid state)**

$= P \times \frac{nRT}{P} = nRT\ (\because P_{H_2} \times V = nRT)$

$= 2 \times 2 \times 300 = 1200\mathrm{cal}$

Q.50 Answer:

—

Solution:

Given,

Specific conductance, $\mathrm{k} = \ 1.826 \times 10^{-6}\ \mathrm{mho}\ \ \mathrm{cm}^{-1}$

Limiting molar conductivity, $\mathring{\Lambda}_{\mathrm{AgCl}} = 138.26\ \mathrm{mho}\ \ \mathrm{cm}^2\ \ \mathrm{mol}^{-1}$

And molar mass of $\mathrm{AgCl} = 143.5\ \mathrm{g}$

We know that,

Solubility, $\mathrm{S} = \frac{\kappa \times 1000}{\mathring{\Lambda}_{\mathrm{AgCl}}}$

Substituting the values,

$\Rightarrow \mathrm{S} = \frac{1.826 \times 10^{-6} \times 1000}{138.26} = 0.0132 \times 10^{-3}\ \mathrm{mol/L}$

Solubility of $\mathrm{AgCl} = 0.0000132 \times 143.5 = 0.00189\ \mathrm{g/L} = 1.89 \times 10^{-3}\ \mathrm{g/L}$

Q.51 Answer:

3

Solution:

The given equation is

$y^3 \cdot \left(3x^2 \cos\left(x^3\right)dx\right) + \sin\left(x^3\right)\left(3y^2 dy\right) = xdx$

$\Rightarrow d\left(\sin\left(x^3\right) \cdot y^3\right) = xdx$

On integrating, we get,

$\sin\left(x^3\right) \cdot y^3 = \frac{x^2}{2} + C$

$\Rightarrow k = 3.$

Q.52 Answer:

235

Solution:

$$x^2 - y^2$$

10 numbers are of the form $3\ \mathrm{K}_1$ **10 numbers are of the form** $3\ \mathrm{K} + 1$

10 numbers are of the form $3\ \mathrm{K} + 2$

$\therefore$ $x^2 - y^2$ **is divided by if we select two number from either** $3\mathrm{k}$ **form or** $3\mathrm{k} + 1$ **form**

Number of ways will be ${}^{10}C_2 + {}^{20}C_2$

$\frac{10\times9}{2} + \frac{20\times19}{2} = 45 + 190 = 235$

Q.53 Answer:

$\frac{\pi}{4}$

Solution:

Let $I = \int_0^{\frac{\pi}{4}} [\sin x + [\cos x + [\tan x + [\sec x]]]] dx$

$\Rightarrow I = \int_0^{\frac{\pi}{4}} [\sin x] dx + \int_0^{\frac{\pi}{4}} [\cos x] dx + \int_0^{\frac{\pi}{4}} [\tan x] dx + \int_0^{\frac{\pi}{4}} [\sec x] dx$

$\Rightarrow I = \int_0^{\frac{\pi}{4}} (0) dx + \int_0^{\frac{\pi}{4}} (0) dx + \int_0^{\frac{\pi}{4}} (0) dx + \int_0^{\frac{\pi}{4}} (1) dx$

$\left\{\because 0 < \sin x,\ \cos x < \frac{1}{\sqrt{2}},\ 0 < \tan x < 1,\ 1 < \sec x < \sqrt{2}\right\}$

$\Rightarrow I = [x]_0^{\frac{\pi}{4}} = \frac{\pi}{4}$

Q.54 Answer:

$\frac{31}{131}$

Solution:

$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = 0.5 \Rightarrow P(A \cap B) = 0.5 \times 0.69 = 0.345$

$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = 0.5$

$\Rightarrow P(A \cap B) = 0.5 \times 0.69 = 0.345$

$P\left(\frac{A}{A^c \cup B^c}\right) = \frac{P[A \cap (A^c \cup B^c)]}{P(A^c \cup B^c)} = \frac{P(A \cap B^c)}{P(A \cap B)^c}$

$= \frac{P(A) - P(A \cap B)}{1 \cdot P(A \cap B)} = \frac{0.5 - 0.345}{1 - 0.345} = \frac{0.155}{0.655} = \frac{31}{131}$

Q.55 Answer:

$\sqrt{\frac{2}{\pi}}$

Solution:

Given limit can be written as

$$L = \lim_{x \to 1^-} \frac{\sqrt{\pi} - \sqrt{2 \sin^{-1} x}}{\sqrt{1 - x}} \times \frac{\sqrt{\pi} + \sqrt{2 \sin^{-1} x}}{\sqrt{\pi} + \sqrt{2 \sin^{-1} x}}$$

$$\Rightarrow L = \lim_{x \to 1^-} \frac{\pi - 2 \sin^{-1} x}{\sqrt{1 - x}\left(\sqrt{\pi} + \sqrt{2 \sin^{-1} x}\right)} = \lim_{x \to 1^-} \frac{2 \cos^{-1} x}{\sqrt{1 - x}\left(\sqrt{\pi} + \sqrt{2 \sin^{-1} x}\right)}$$

$\Rightarrow L = \lim_{x \to 1^-} \frac{\pi - 2\sin^{-1} x}{\sqrt{1-x}\left(\sqrt{\pi} + \sqrt{2\sin^{-1} x}\right)} = \lim_{x \to 1^-} \frac{2\cos^{-1} x}{\sqrt{1-x}\left(\sqrt{\pi} + \sqrt{2\sin^{-1} x}\right)}$

Let $K = \lim_{x \to 1^-} \frac{\cos^{-1} x}{\sqrt{1-x}}$ **and put** $x = \cos\theta$**, we get**

$$K = \lim_{\theta \to 0} \frac{\frac{\theta}{2} \cdot 2}{\sqrt{2} \cdot \sin\frac{\theta}{2}} = \lim_{\theta \to 0} \frac{\sqrt{2}}{\frac{\sin\frac{\theta}{2}}{\frac{\theta}{2}}} = \sqrt{2}$$

$$\therefore L = \frac{2\sqrt{2}}{2\sqrt{\pi}} = \sqrt{\frac{2}{\pi}}.$$

$\therefore L = \frac{2\sqrt{2}}{2\sqrt{\pi}} = \sqrt{\frac{2}{\pi}}.$

Q.56 Answer:

-1

Solution:

Given, $\mathrm{Z} = \mathrm{PQ}^{-1}$

$|\mathrm{P}| = -5,\ |\mathrm{Z}| = 10$

$|\mathrm{Z}| = \frac{|\mathrm{P}|}{|\mathrm{Q}|} \Rightarrow |\mathrm{Q}| = \frac{-5}{10} = -\frac{1}{2}$

$Z|\mathrm{Q}| = \mathrm{P.\,adj(Q)} \Rightarrow \mathrm{P.\,adj(Q)} = -\frac{1}{2}\mathrm{Z}$

$\mathrm{Tr((adjQ).P)} = \mathrm{Tr(P.adjQ)}$

$= \left(-\frac{1}{2}\mathrm{Tr(Z)}\right)$

$= -1$

Q.57 Answer:

81

Solution:

$n_1 = 40, n_2 = 50$

$\overline{x}_1 = 54, \overline{x}_2 = 63$

$\sigma_1 = 6, \sigma_2 = 9$

$\overline{x} = \frac{n_1\overline{x}_1 + n_2\overline{x}_2}{n_1+n_2} = \frac{40\times54+50\times63}{90} = 59$

$d_1 = \overline{x} - \overline{x}_1 = 5$

$d_2 = \overline{x}_2 - \overline{x} = 4$

Combined $\text{S. D} = \sqrt{\frac{n_1d_1^2+n_2d_2^2+n_1\sigma_1^2+n_2\sigma_2^2}{n_1+n_2}}$

$= \sqrt{\frac{40\times25+50\times16+40\times36+50\times81}{90}}$

$= 9$

Variance $= 81$

Q.58 Answer:

None of these

Solution:

If " $a+b$ " is a root it satisfies the equation

Hence $(a+b)^2 + a(a+b) + b = 0$

$\Rightarrow \quad 2a^2 + 3ba + \left(b^2+b\right) = 0$

Now since "a" is an integer Discriminant is a perfect square

$\Rightarrow \quad 9b^2 - 8\left(b^2+b\right) = p^2 \text{ for same } p \in Z$

$\Rightarrow \quad (b-4)^2 - 16 = p^2$

$\Rightarrow \quad (b-4+b)(b-4-p) = 16$

$b-4+p = \pm8, \quad b-4-p = \pm2, \quad b-4+p = b-4-p = \pm4$

So $\quad b-4 = 5, -5, 4, -4$

$\Rightarrow \quad b = 9, -1, 8, 0 \Rightarrow \quad \left(b^2\right)_{\max} = 81$

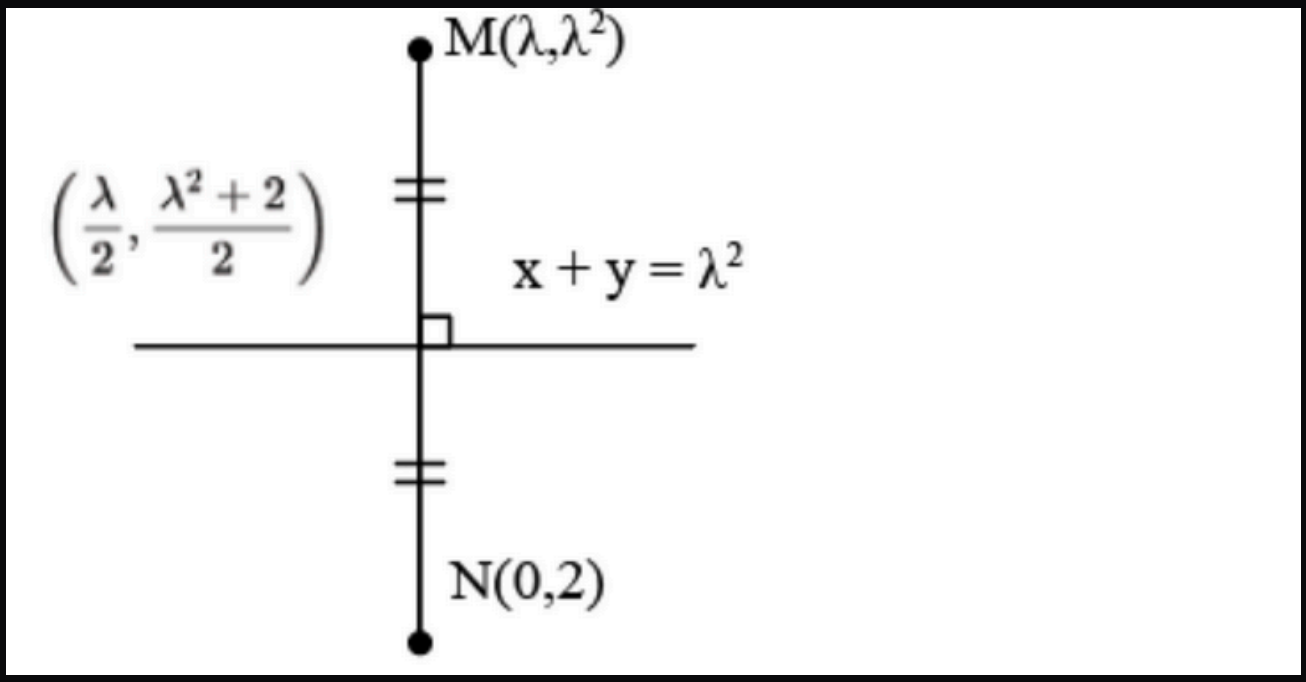
Q.59 Answer:

5

Solution:

Since the mid-point of the line segment MN **lie on the given line,** $\Rightarrow \frac{\lambda}{2} + \frac{\lambda^2+2}{2} = \lambda^2 \ldots .$ **(i)**

Also, the line segment MN **is perpendicular to the given line** $\Rightarrow \left(\frac{\lambda^2-2}{\lambda-0}\right) \times (-1) = -1 \ldots \ldots .$ **(ii)**



As (i) and (ii) must be satisfied simultaneously, ∴ On solving, we get,

$$\lambda = -1, 2$$
$$\Rightarrow \lambda_1^2 + \lambda_2^2 = 1 + 4 = 5$$

Q.60 Answer:

$x \in \left[1, 1+\sqrt{2}\right]$

Solution:

(i) $-1 \leq 2x - x^2 \leq 1$ **(for** $\sin^{-1}$ **to be defined)**

$\Rightarrow -1 \leq x^2 - 2x \leq 1$

i.e. $x^2 - 2x + 1 \geq 0$ **and** $x^2 - 2x - 1 \leq 0$

$(x-1)^2 \geq 0$ **and** $(x-1)^2 - \left(\sqrt{2}\right)^2 \leq 0$

$x \in R$ **and** $\left(x - 1 - \sqrt{2}\right)\left(x - 1 + \sqrt{2}\right) \leq 0$

$\Rightarrow x \in \left[1-\sqrt{2}, 1+\sqrt{2}\right] \ldots (1)$

(ii) $2 - \frac{1}{|x|} \geq 0 \Rightarrow \frac{1}{|x|} \leq 2 \Rightarrow |x| \geq \frac{1}{2} \Rightarrow x \in \left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right) \ldots (2)$

(iii) $\left[x^2\right] \neq 0 \Rightarrow x^2 \notin [0,1)$

$\Rightarrow x \notin (-1,1) \Rightarrow x \in (-\infty, -1] \cup [1, \infty) \ldots (3)$

Hence, $(1) \cap (2) \cap (3)$

$\Rightarrow x \in \left[1, 1+\sqrt{2}\right]$

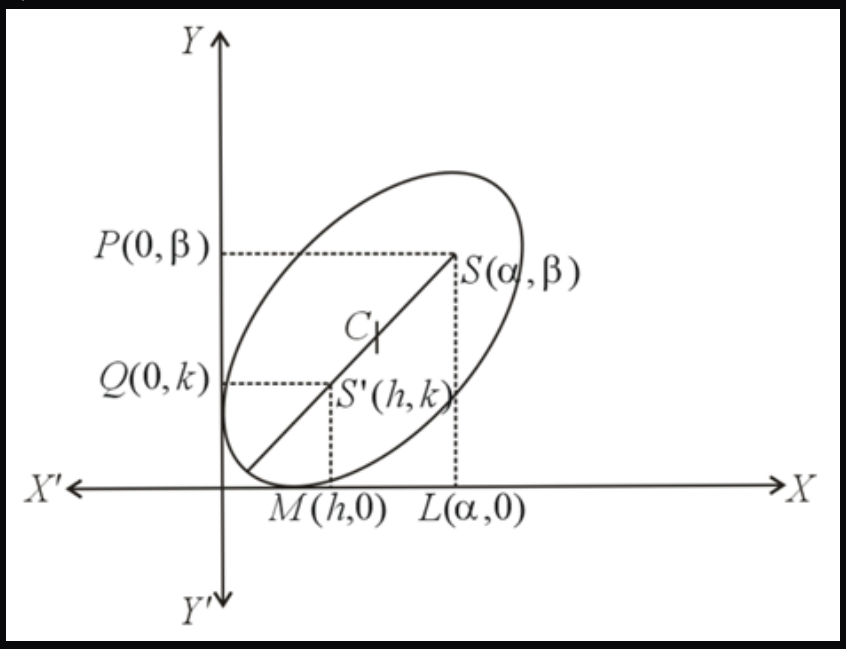
Q.61 Answer:

$\left(x^2 + y^2\right)\left(x^2y^2 + b^4\right) = 4a^2x^2y^2$

Solution:

Let $S(\alpha, \beta)$ and $S'(h, k)$ be the foci of the ellipse. We know that the product of the perpendiculars from the foci of an ellipse to any tangent is equal to the square of semi-minor axes. Therefore,

$k\beta = b^2$ and $h\alpha = b^2$



Now,

$SS' = 2ae$

$\Rightarrow (\alpha - h)^2 + (\beta - k)^2 = 4a^2e^2$

$\Rightarrow \left(\alpha - \frac{b^2}{\alpha}\right)^2 + \left(\beta - \frac{b^2}{\beta}\right)^2 = 4a^2e^2$

$\Rightarrow \frac{(\alpha^2 - b^2)^2}{\alpha^2} + \frac{(\beta^2 - b^2)^2}{\beta^2} = 4a^2e^2$

$\Rightarrow \beta^2(\alpha^2 - b^2)^2 + \alpha^2(\beta^2 - b^2)^2 = 4a^2\alpha^2\beta^2e^2$

$\Rightarrow (\alpha^2 + \beta^2)b^4 + \beta^2\alpha^4 + \alpha^2\beta^4 - 4\alpha^2\beta^2b^2 = 4a^2\alpha^2\beta^2\left(\frac{a^2 - b^2}{a^2}\right)$

$\Rightarrow (\alpha^2 + \beta^2)b^4 + \alpha^2\beta^2(\alpha^2 + \beta^2) = 4\alpha^2\beta^2(b^2 + a^2 - b^2)$

$\Rightarrow (\alpha^2 + \beta^2)(\alpha^2\beta^2 + b^4) = 4a^2\alpha^2\beta^2$

Hence, the locus of $(\alpha, \beta)$is

$(x^2 + y^2)(x^2y^2 + b^4) = 4a^2x^2y^2$.

Similarly, by eliminating α and β, we can show that the locus of $S'(h, k)$ is $(x^2 + y^2)(x^2y^2 + b^4) = 4a^2x^2y^2$.

Q.62 Answer:

0

Solution:

$z^2 - 4z = |z|^2 + \frac{16}{|z|^3}$

$x^2 - y^2 - 2ixy - 4x - 4iy = |z|^2 + \frac{16}{|z|^3} = \text{real}$

Since, R. H. S. is a real number so L. H. S. must also be real. So, comparing both sides

$\Rightarrow -2xy - 4y = 0 \Rightarrow y(x - 2) = 0$

But $x \neq 2$, so $y = 0 \Rightarrow z = x$

$\therefore \quad x^2 = 4x + x^2 + \frac{16}{x^3}$

$\Rightarrow \quad x^4 = -4$, which is not possible. So number of solutions is zero.

Q.63 Answer:

9

Solution:

7th term from the beginning is ${}^nC_6(2)^{\frac{n-6}{3}}\left(\frac{1}{3}\right)^{\frac{6}{3}}$

7th term from the end is ${}^nC_6\left(\frac{1}{3}\right)^{\frac{n-6}{3}}(2)^{\frac{6}{3}}$

$\Rightarrow \frac{{}^nC_6(2)^{\frac{n}{3}-2}\left(\frac{1}{3}\right)^2}{{}^nC_6 2^2\left(\frac{1}{3}\right)^{\frac{n}{3}-2}} = \frac{1}{6}$

$\Rightarrow (2)^{\frac{n}{3}-4}\left(\frac{1}{3}\right)^{4-\frac{n}{3}} = \frac{1}{6}$

$\Rightarrow (2\times 3)^{\frac{n}{3}-4} = (2\times 3)^{-1}$

$\Rightarrow n = 9$

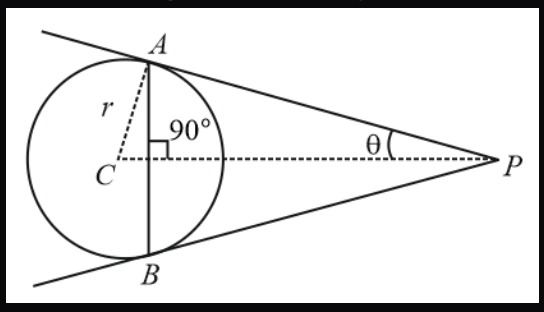
Q.64 Answer:

8

Solution:

Given, $\frac{1}{r^2}+\frac{1}{PA^2}=\frac{1}{16}$

From the diagram, we can say that, $\tan\theta = \frac{r}{PA}$



Now substitute value of r **in the given equation,**

$\Rightarrow \frac{\cot^2\theta+1}{(PA)^2}=\frac{1}{16}$

$\Rightarrow \frac{\operatorname{cosec}^2\theta}{PA^2}=\frac{1}{16}$

$\Rightarrow PA\sin\theta = 4 \ \ldots(\text{i})$

Now from the figure,

$\sin\theta = \frac{x}{PA} \Rightarrow 2x = 8 = AB$ **(using** (i)**)**

Q.65 Answer:

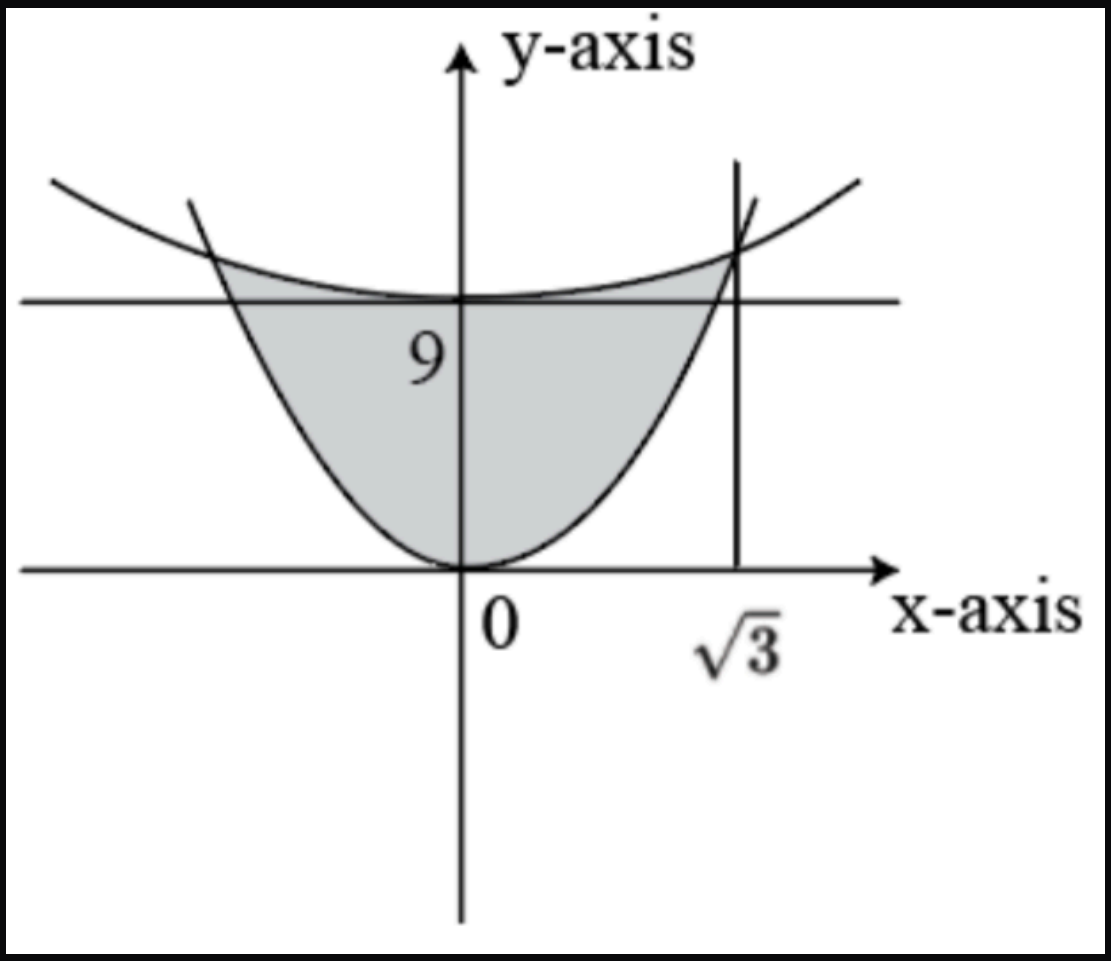
$12\sqrt{3}$

Solution:

$x^2 = \frac{y}{5}, x^2 = \frac{1}{2}(y-9)$

Area $= 2\int_0^{\sqrt{3}}\left[\left(2x^2+9\right)-\left(5x^2\right)\right]dx$

$= 2\int_0^{\sqrt{3}}\left(9-3x^2\right)dx = 18x-2x^3 = 18\sqrt{3}-6\sqrt{3} = 12\sqrt{3}$



Q.66 Answer:

$\frac{1}{3}(-2\hat{i}-\hat{j}+2\hat{k})$

Solution:

We have

$$\vec{r}\times\vec{a}+\vec{r}\times\vec{b}=\vec{a}\times\vec{b}+\vec{b}\times\vec{a}=\vec{0}$$

$$\Rightarrow \vec{r}\times(\vec{a}+\vec{b})=\vec{0}$$

$$\Rightarrow \vec{r}=\lambda(\vec{a}+\vec{b})=\lambda(-2\hat{i}-\hat{j}+2\hat{k})$$

$$\Rightarrow \hat{r}=\frac{1}{3}(-2\hat{i}-\hat{j}+2\hat{k})$$

Q.67 Answer:

All of these

Solution:

$A88,\ 6B8,\ 86C$ **is divisible by 72.**

Assume $A88 = 72a,\ \ 6B8 = 72b,\ \ 86C = 72c$

Given

$$\begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ 8 & 8 & C \end{vmatrix}$$

$R_3 \Leftrightarrow 100R_1 + 10R_2 + R_3$

$$\Rightarrow \begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ 8 & 8 & C \end{vmatrix} = \begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ A88 & 6B8 & 86C \end{vmatrix} = 72 \begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ a & b & c \end{vmatrix}$$

which is divisible by 72.

for A88, 6B8, 86C to be individually divisible by 72, we get a value of A=2, B=4, C=4.

Hence, determinant is divisible by both 144 and 288 as well.

Q.68 Answer:

$\frac{1}{3}$

Solution:

$\tan\theta = 3\tan\phi$

$\tan(\theta - \phi) = \frac{\tan\theta - \tan\phi}{1 + \tan\theta\tan\phi}$

$= \frac{2\tan\phi}{1 + 3\tan^2\phi} = \frac{2}{\cot\phi + 3\tan\phi}$

Using $\mathrm{AM} \geq \mathrm{GM}$

$\frac{\cot\phi + 3\tan\phi}{2} \geq \sqrt{3}$

$\Rightarrow \frac{2}{\cot\phi + 3\tan\phi} \leq \frac{1}{\sqrt{3}}$

$\Rightarrow \tan(\theta - \phi) \leq \frac{1}{\sqrt{3}}$

$\Rightarrow \tan^2(\theta - \phi) \leq \frac{1}{3}$

Q.69 Answer:

$\mathrm{g(x)}$ is an increasing function

Solution:

$\ell\mathrm{n}\ \mathrm{f(x)} > 0 \Rightarrow \mathrm{f(x)} > 1$

$f(x)$ **...(i) is second degree polynomial and coefficient of** x^2 **is positive and, and** $f'(x)$ **is a linear expression**

$f'(x) + 21f(x)$ **...(ii) is aso second degree polynomial**

eqn (i) and (ii) have same coefficient of x^2 **and eqn (ii) doest not have any real root hence it should be greater than zero**

$\mathrm{f\prime\ (x) + 21\ f(x) > 0\ \forall\ x \in R}$ **...(iii)**

$g(x) = e^{21x} f(x)$

$\mathrm{g'(x) = e^{21x}\ f\prime\ (x) + 21\ e^{21x} f(x)}$

From (iii), $g'(x) > 0$

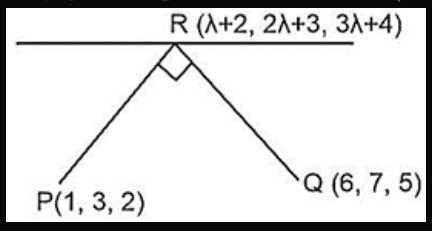
$\Rightarrow$ **g(x) is an increasing function**

Q.70 Answer:

45

Solution:

Any general point on the line is $(\lambda+2, 2\lambda+3, \lambda+4)$



$\overrightarrow{PR} =< \lambda+1, 2\lambda, 3\lambda+2 >$

$\overrightarrow{RQ} =< 4-\lambda, 4-2\lambda, 1-3\lambda >$

From the diagram, $\overrightarrow{PR} \cdot \overrightarrow{RQ} = 0$

$\Rightarrow (\lambda+1)(4-\lambda) + 2\lambda(4-2\lambda) + (3\lambda+2)(1-3\lambda) = 0$

$\Rightarrow \lambda^2 - 3\lambda - 4 + 4\lambda^2 - 8\lambda + 9\lambda^2 + 3\lambda - 2 = 0$

$\Rightarrow 14\lambda^2 - 8\lambda - 6 = 0$

$\Rightarrow \lambda = 1, \ -\frac{3}{7}$

Coordinates of R **are** $(3, 5, 7)$ **or** $\left(\frac{11}{7}, \frac{15}{7}, \frac{19}{7}\right)$

Least value of $7(a+b+c) = 11 + 15 + 19 = 45$

Q.71 Answer:

—

Solution:

Let $x = t^4 \Rightarrow dx = 4t^3 dt$

$\Rightarrow I = \int_0^2 \frac{t \cdot 4t^3 dt}{1+t^2}$

$\Rightarrow \frac{I}{4} = \int_0^2 \frac{t^4-1+1}{1+t^2} dt$

$\Rightarrow \frac{I}{4} = \int_0^2 \left(\frac{(t^2-1)(t^2+1)}{t^2+1} + \frac{1}{t^2+1} \right) dt$

$= \left[\frac{t^3}{3} - t + \tan^{-1} t \right]_0^2$

$\Rightarrow \frac{I}{4} = \frac{8}{3} - 2 + \tan^{-1} 2$

$\therefore I = \frac{8}{3} + 4\tan^{-1} 2$

$\Rightarrow k = \frac{8}{3}, m = 2$

$\therefore 3k - m = 6$

Q.72 Answer:

—

Solution:

Let $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots a_n$

Putting in given equation and comparing

Coefficients of powers of x,

We get $f(x) = x + b$ **. Where b is constant**

$\therefore \quad f(9) - f(3) = 6$

Q.73 Answer:

—

Solution:

Eliminating n between the given relations, we get

$\ell^2 + \mathrm{m}^2 - (-\ell - \mathrm{m})^2 = 0$

or $\ell\mathrm{m} = 0 \;\Rightarrow\; \ell = 0 \text{ orm} = 0$

Now $\ell = 0$

$\Rightarrow \quad 1.\ell + 0.\mathrm{m} + 0.n = 0 \text{ and } \ell + \mathrm{m} + n = 0$

60

Solving these we get

$\frac{\ell}{0} = \frac{\mathrm{m}}{-1} = \frac{n}{1}$ **....(i)**

Similarly $\mathrm{m} = 0 \Rightarrow \quad 0.\ell + 1.\mathrm{m} + 0.\mathrm{n} = 0$ and $\ell + \mathrm{m} + \mathrm{n} = 0$

Solving these we get $\frac{\ell}{1} = \frac{\mathrm{m}}{0} = \frac{\mathrm{n}}{-1}$ ..... (ii)

thus dr's of given lines are 0, -1, 1 and 1, 0, -1

respectively.

Let θ **is angle between them**

$\cos\theta = \frac{0.1+(-1).0+1(-1)}{\sqrt{0^2+(-1)^2+1^2}\sqrt{1^2+0^2+(-1)^2}}$ **= 1/2**

Q.74 Answer:

—

Solution:

$$15 < x_1 + x_2 + x_3 \leq 20$$
$$\Rightarrow x_1 + x_2 + x_3 = 16 + r, \text{ where } r = 0, 1, 2, 3, 4$$

Now, the number of positive integral solutions of $x_1 + x_2 + x_3 = 16 + r$ is $^{16+r-1}C_{3-1} = {}^{15+r}C_2$

Thus, the required number of solutions is

$$\begin{aligned}
&\sum_{r=0}^{4} {}^{15+r}C_2 = {}^{15}C_2 + {}^{16}C_2 + {}^{17}C_2 + {}^{18}C_2 + {}^{19}C_2 \\
&= {}^{15}C_2 - {}^{16}C_3 + \left({}^{16}C_3 + {}^{16}C_2 + {}^{17}C_2 + {}^{18}C_2 + {}^{19}C_2\right) \\
&= {}^{15}C_2 - {}^{16}C_3 + {}^{20}C_3 = 685
\end{aligned}$$

Q.75 Answer:

—

Solution:

$$\begin{aligned}
&A_1^T = -A_1, A_2^T = -A_2, \ldots, A_{20}^T = -A_{20} \\
&B = \sum_{r=1}^{20} 2r \left(A_r\right)^{2r+1} \\
&B^T = \left(\sum_{r=1}^{20} 2r \left(A_r\right)^{2r+1}\right)^T \\
&= \sum_{r=1}^{20} 2r \left(A_r^T\right)^{2r+1} \\
&= \sum_{r=1}^{20} 2r \left(-A_r\right)^{2r+1} \\
&= -\sum_{r=1}^{20} 2r \left(A_r\right)^{2r+1} \\
&= -B \\
&\Rightarrow B \text{ is skew-symmetric}
\end{aligned}$$

Hence, the sum of principal diagonal elements of $B = 0$